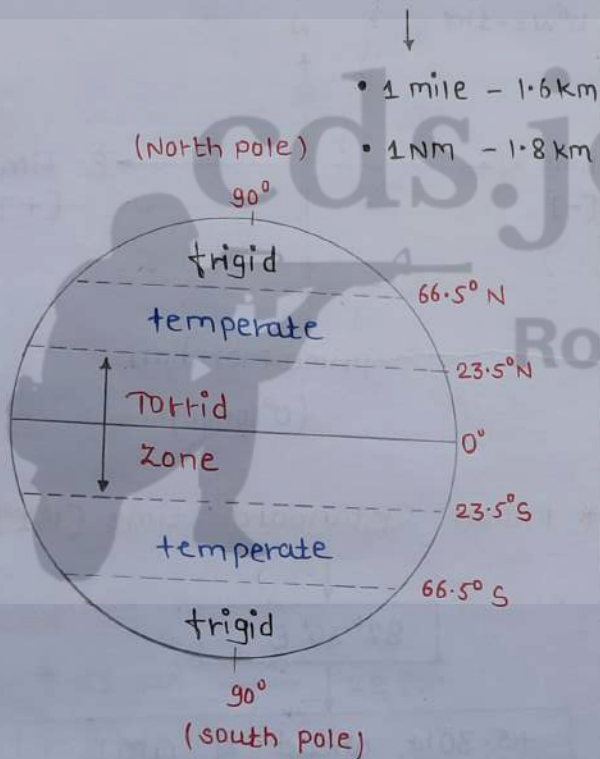


PHYSICAL GEOGRAPHY

Latitude

- Imaginary Horizontal Lines.
- that run from East to West
- Latitude - "parallels"
- ↓
- Total - 181 (90°N , 90°S , 0°)
- $1^{\circ}\text{Lat} = 111\text{ km}$ (69 miles)



* Important Latitude

- 0° - Equator
- 23.5°N - Tropic of cancer
- 23.5°S - Tropic of capricorn
- 66.5°N - Arctic circle
- 66.5°S - Antarctic circle.

- largest Latitude - Equator
- Smallest Latitude - poles (N/S)

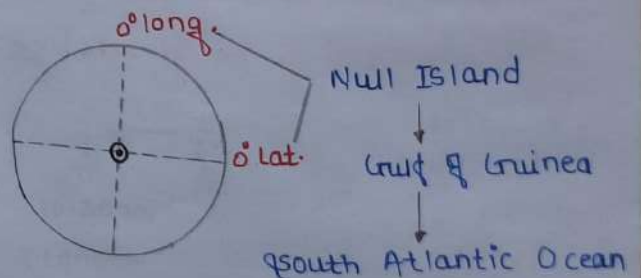
Longitudes

- Imaginary verticle lines
- ↓
- North to south
- Total Longitudes - 360 (180°E & 180°W)
- ↓
- 180°W & $180^{\circ}\text{E} = 1\text{ longitude}$
- 0°long. - Prime meridian
- $180^{\circ}\text{meridian}$ - International date line.
- Earth rotates 360° in 24 hours

$$1\text{ hr} = \frac{360}{24} = 15^{\circ}$$

$$1^{\circ} = 4\text{ min}$$

- All meridians of longitudes are of equal length.



* Longitudes prime meridian

❖ It passes through Greenwich in London.

❖ TRICK - BSF GAME in TAGO Kingdom.

* 0° Long.

- UK
- France
- Spain
- Algeria
- Mali
- Burkina Faso
- Faso
- Togo
- Ghana

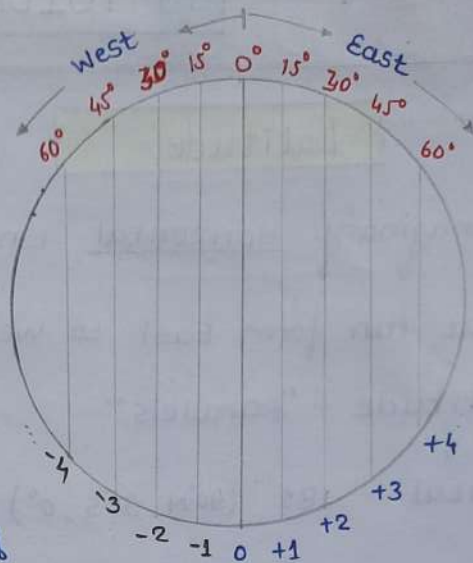
* Why the length of days on equator is always more than Night.

⇒ Refraction

* If there was no atmosphere.

↓
No refraction

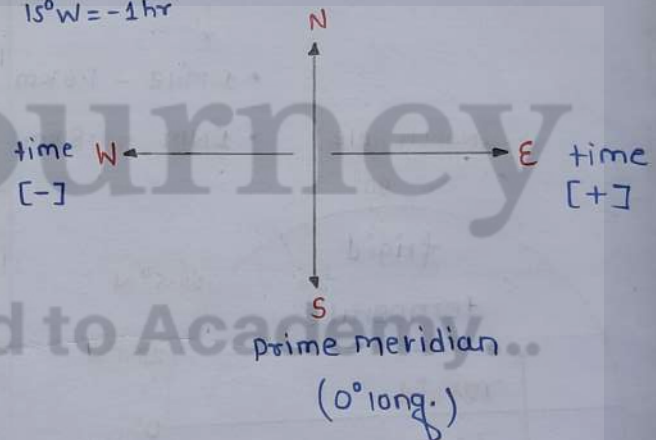
↓
day = Night



• every

$15^\circ E = +1 \text{ hr}$

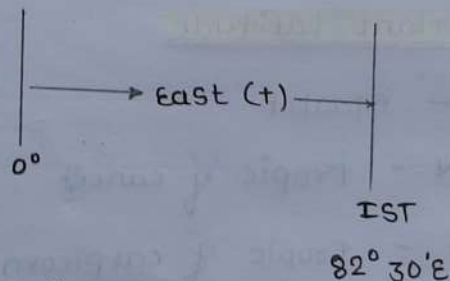
$15^\circ W = -1 \text{ hr}$



* Indian Standard time [IST]

↓
 $82^\circ 30' E$

↓
+5.30hr ahead of GMT



• $15^\circ = 1 \text{ hr}$

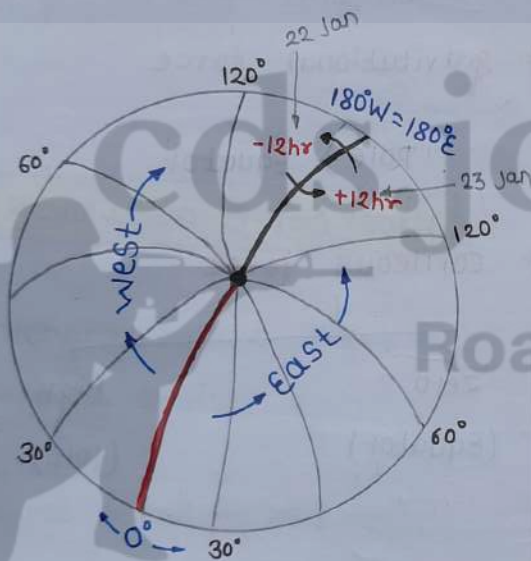
• $82^\circ 30' E = +5.30^\circ$

* IST - $82^{\circ} 30' E$

↓
passes through

- UP
- MP
- Chatt.
- Odisha
- A.P

* chaibagan time - 1hr ahead of IST.



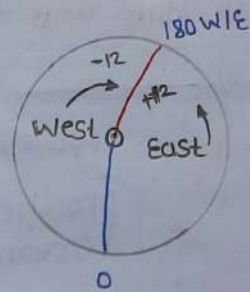
Example:

* 23 Jan → 22 Jan

- gain day / time lose (24hr)

* 22 Jan → 23 Jan

- time ~~lose~~ / Lose day gain



Toward East

- gain day
- lose time

Toward West

- lose day
- gain time

West to East

- gain a day
- Ex: 24 → 23

East to West

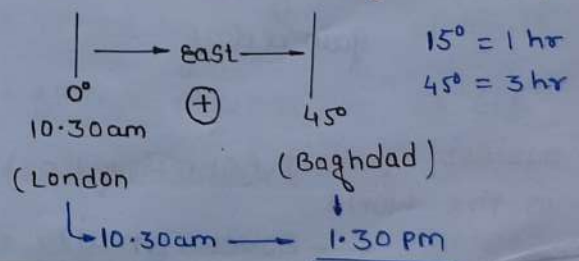
- lose a day
- Ex: 23 → 24

* 24 hr lose = 1 day gain

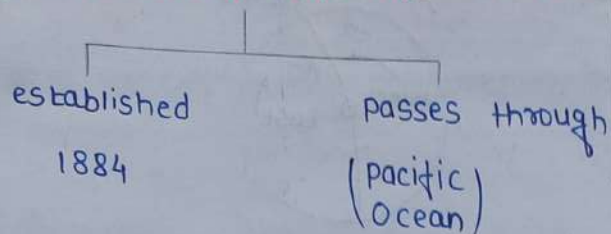
Q.1 What will happen if you are flying east across the IDL?

→ you will lose 24 hours

Q.2 If News is broadcast from London at 10:30am at what time it will be heard in Baghdad ($45^{\circ} E$)



* International date line - 180° E/W



* Which Strait is nearest to IDL

- | | | |
|-----------------------|-----------------------|---------------------|
| ① Malacca Strait | Arctic Ocean (Russia) | Pacific Ocean (USA) |
| ✓ ② Bering Strait | | |
| ③ Strait of Florida | | |
| ④ Strait of Gibraltar | | |

* Time zones

- France - 13
- Russia - 11
- US - 6
- South A. - 2

* IDL $\rightarrow$ passes water body only.

Q. When a ship crosses the IDL from West to East

$\Rightarrow$ moves forward by one day.
 $\downarrow$
gain a day

Q. busiest Strait (shipping line) in the world
 $\rightarrow$ Dover Strait.

Shape of the earth

Geoid or oblate spheroid

Reason $\rightarrow$ More centrifugal force at equator.

* Radius of the earth

Equatorial radius > Polar radius

* gravitational force

pole > Equator

* Coriolis force

Zero (Equator) high (pole)

* Speed of earth (Rotational)

max. (equator) min. (pole)

* Day & Night

Rotation of earth

Season

Revolution on tilted axis

(Earth revolution and inclination of its axis)

Rotation of the earth

Rotation - Spinning on its own axis.

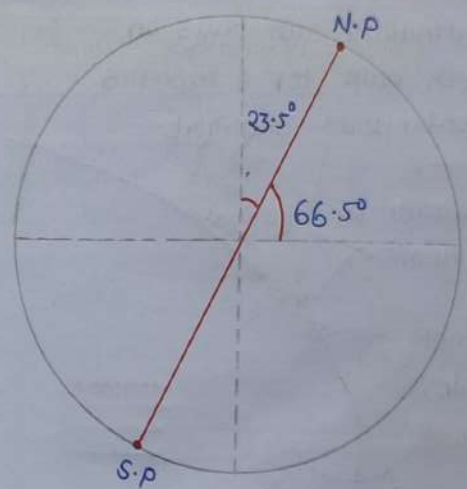
one - 23hr, 56m, 4sec

Direction - West (to) east

Rotational speed is

→ max - at Equator

→ min - at pole



• Axial Inclination - 23.5°

• orbital inclination - 66.5°

* due to tiltiness of earth
↓
changes of season

Revolution of the earth

Revolving around the sun in Elliptical orbit.

orbital speed - 29.8 km/s

max. (Mercury) min (Neptune)

Axis

Rotation

own axis

speed

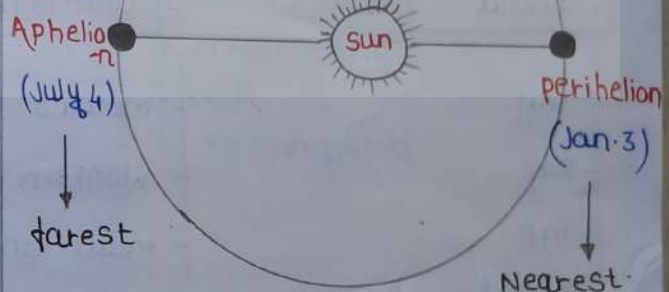
23hr 56m 4sec.

orbit

Revolution

speed

365 Day 6 hr, 9min 9sec.



• perihelion - 3 Jan.

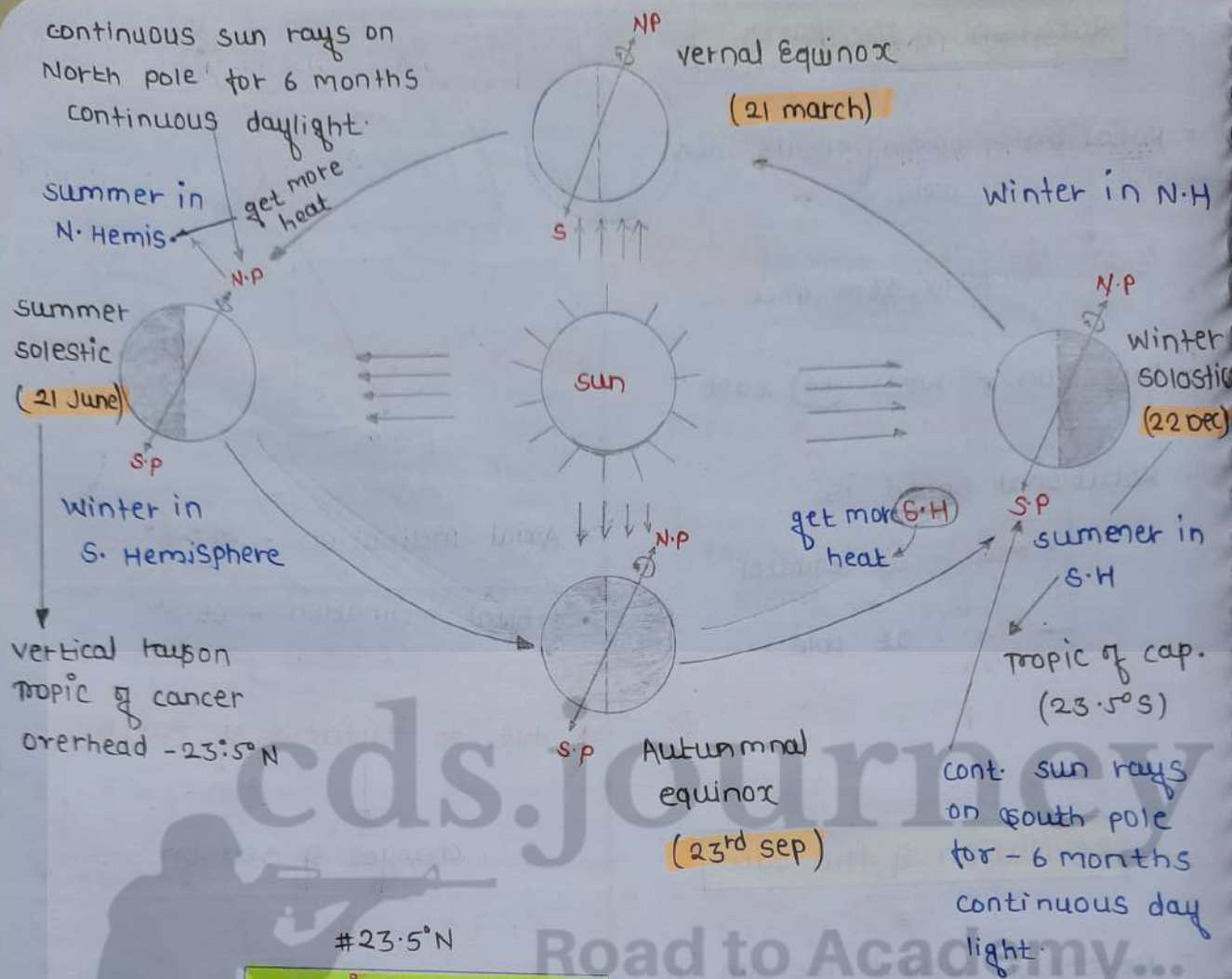
• Aphelion - 4 July

• vernal equinox - 21 march

• Autumnal equinox - 23 sep.

• summer solstice - 21 Jan

• winter solstice - 21 Dec



Tropic of cancer		
India	countries	
Guj.	Mexico	Mexico Se Mumbai Me Aane ke Kiye Egg should Bring to my china Town.
Raj.	Western Sahara	India
MP	Mauritania	Bangladesh
Chhat.	Mali	Myanmar
Jharkhand	Algeria	China
W.B	Libya	Taiwan
Tripura	Egypt	
Mizoram	Saudi Arabia	
	UAE	
	Oman	

Equator Countries

Trick - ECB UK-CGS I

Ecuador

Colombia

Brazil

Gabon

Congo

Democratic Republic of Congo

Uganda

Kenya

Samalia

Maldives

Indonesia

Tropic of Capricorn

Trick - CAB PNB Se Mumbai
Aai

Chile

Argentina

Brazil

Paraguay

Namibia

Botswana

South Africa

Mozambique

Madagascar

Australia

* Five landlocked states

Chhattisgarh

Haryana

Jharkhand

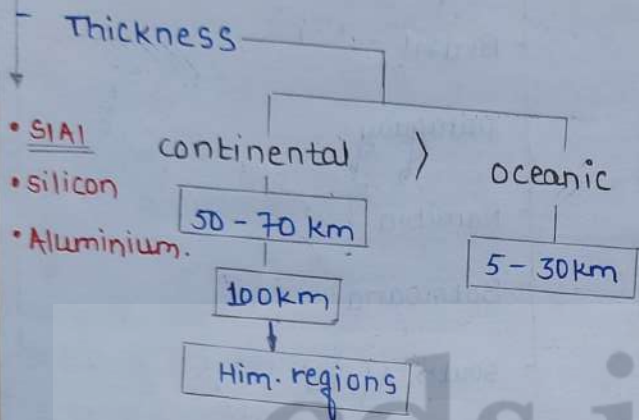
MP

Telangana

Interior of the Earth

Crust

- outermost solid part
- brittle in nature



Mantle

→ 83% of earth

- beyond the crust
• silicon
• magnesium
- extends - depth - 2900 km
- beyond asthenosphere solid mantle

Core

→ 16% of earth

- core made - Nickel & Iron
- nife layer

outer core

- liquid state

inner core

solid state

↓
5100 km

Lithosphere

↓
crust + uppermost mantle.

Asthenosphere

- upper portion of mantle
- extending - 400 km
- main source of magma

↓
find its way to the surface during volcanic eruptions.

- plate tectonic movement
- weaker zone

* Earth crust

- oxygen
- silicon
- Aluminium

* Earth - most Abundant element

- Iron
- oxygen
- silicon.

* Gold mine → 3-4 km

↓
South Africa

* deepest drill at Kola - Arctic ocean

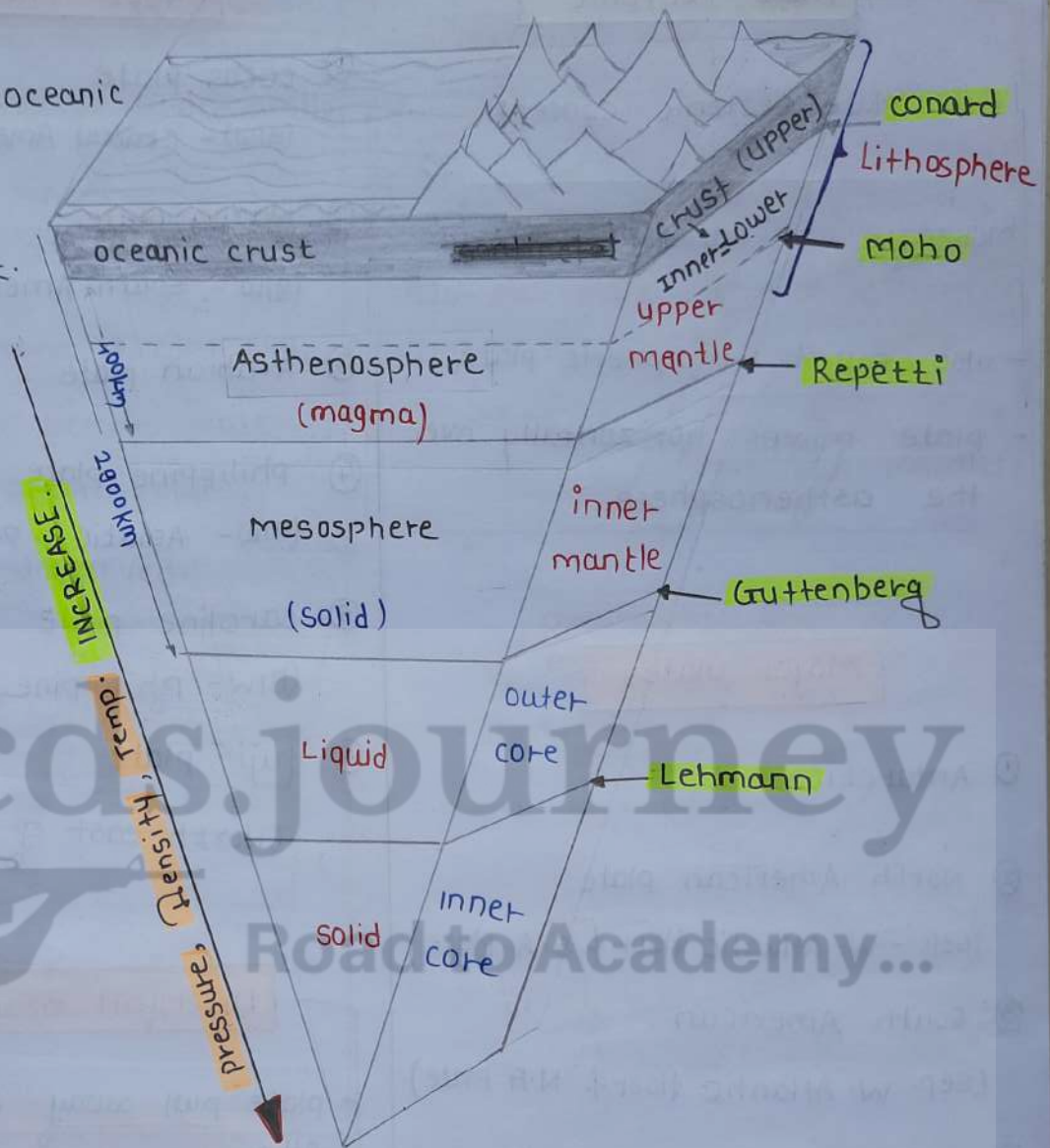
↓
12 km.

* Thickness

continental > oceanic

* denser

oceanic > cont.



* Inner most part of the earth.

* Methods to know Earth's interior

① conard dis.

② moho dis.

③ Guttenberg dis.

④ Lehmann dis.

* Density

core > mantle > crust

Direct source

- volcano
- mining
- Drilling
- Surface
- Rock

Indirect source

- Earthquake
- meteorites
- Gravitational force.

plate tectonic

plate tectonic concept

mckenzie

parker

morgan

- also called lithospheric plate
- plate moves horizontally over the asthenosphere.

Major plates

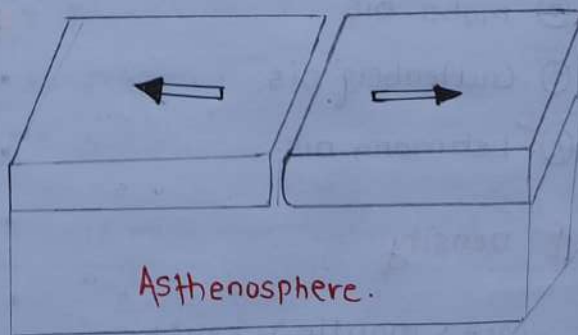
- ① Antarctica
- ② North American plate
(sep. - w. Atlantic floor & S.A. plate)
- ③ South American
(sep. w. Atlantic floor & N.A. plate)
- ④ Pacific plate
- ⑤ India - Aus - NZ. plate
- ⑥ Africa with eastern Atlantic
- ⑦ Eurasia

Minor plates

- ① Cocos plate
(Blw - central America & P.P.)
- ② Nazca plate
(Blw - South America & P.P.)
- ③ Arabian plate
- ④ Philippine plate
(Blw - Asiatic & P.P.)
- ⑤ Caroline plate
(Blw - Philippine & Indian P.)
- ⑥ Fuji plate
(North east of Aus.)

Divergent Boundaries

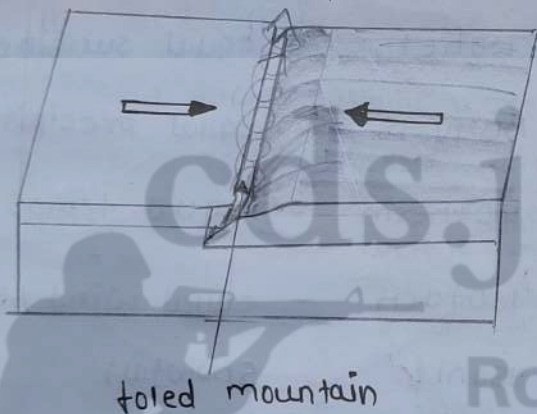
- plate pull away from each other.
- spreading sites.
- Ex. mid-Atlantic Ridge.



① crust generated

Convergent Boundaries

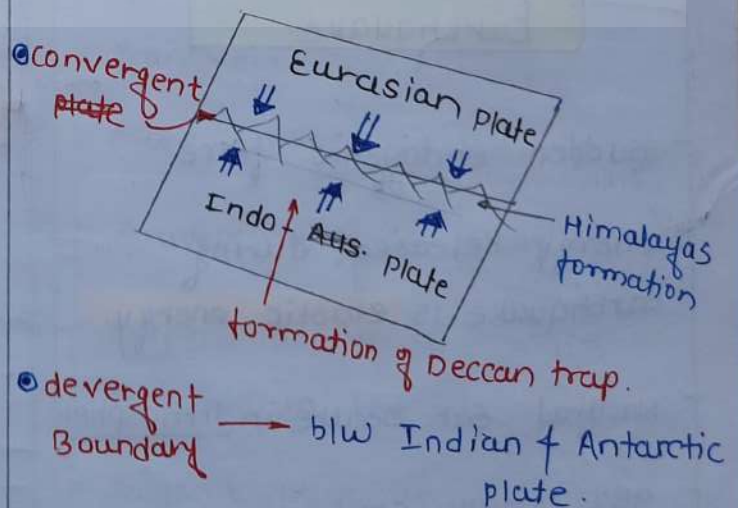
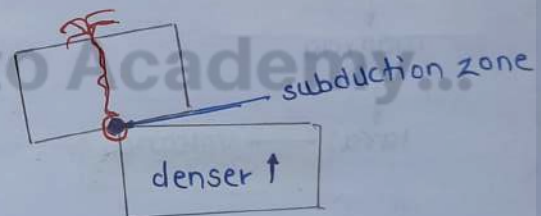
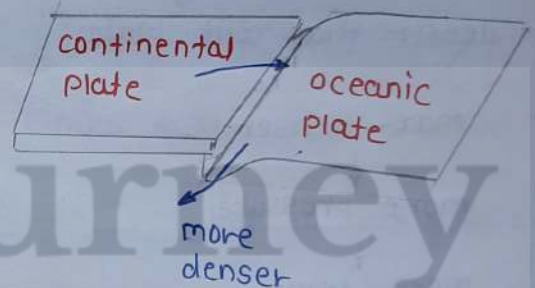
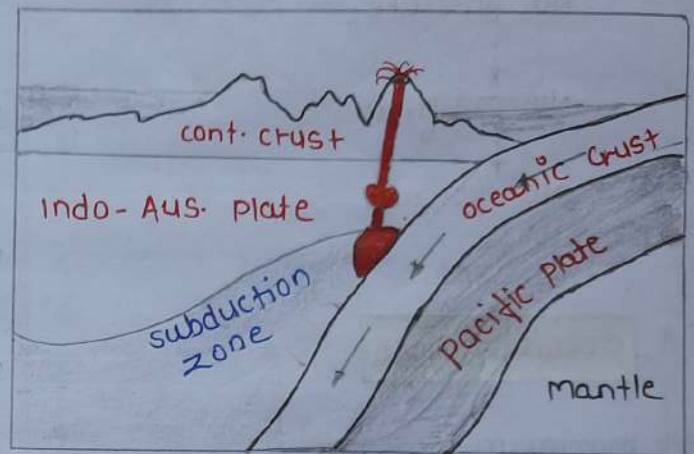
- one plate dived under another
- subduction zone: where sinking of a plate.
- b/w
 - ① oceanic + cont. plate
 - ② two oceanic plate.
 - ③ two cont. plate.
- ⊙ crust is destroyed.



Transform Boundaries

- crust neither produced nor destroyed.
- No formation
- eruption do not takes
- ⊙ rotation of the earth has its effect on the separated blocks of the plate portion.

* oceanic crust is denser than continental crust.



⊙ divergent Boundary → b/w Indian & Antarctic plate.

* lower density → not much heat up.
 ↓
 magma not

* No water
 ↓
 No pressure
 ↓
 No magma
 ↓
 No volcano

* oceanic plate

- maximum volcano
- denser than cont. plate

max. denser
 ↓
 more pressure
 ↓
 more temp.
 ↓
 magma
 ↓
 lava → volcano.

Earthquake

- sudden endogenic force
- energy released during earthquake is **elastic energy**.
- Natural Ear. occur in lithosphere
- 98% oceanic crust
- 2% continental crust

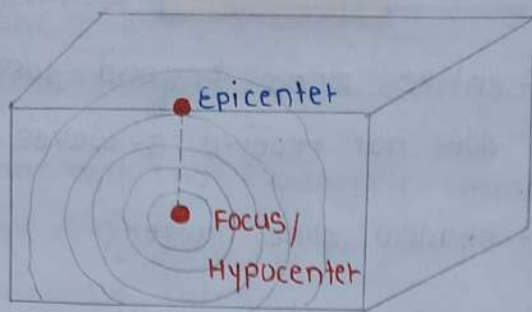
- seismology - study of earthquake
- seismograph - measure seismic waves.
- Isoseismic line - noting equal intensity
- isobar - same pressure
- isohaline - equal salinity
- isohume - equal humidity
- isohel - equal sunshine
- isohyets - equal precipit
- isotherm - equal temp.
- isotach - equal wind speed
- isonit - snowfall

* Focus / Hypocenter:

- origin of earthquake
- point where energy released.

* Epicenter

- point on surface nearest to the focus.
- 1st one to experience the waves.
- point directly above the focus.
- maximum damage.



two types

body wave

surface wave

- generated due to release of energy at focus & move in all direction. ~~travelling~~
- body waves interact with the surface rock & generate new set of waves.
- P-waves
- S-waves
- L-wave

* The velocity of wave changes as they travel through materials with different densities.

* denser material

↓
higher is the velocity.

P-Wave

- move faster
 - 1st to arrive at surface
 - similar to sound waves.
 - travel through
 - gaseous
 - liquid
 - solid
- $S > L > G$
↑
max. velocity
- longitudinal wave (sound)

- 1.7 times faster than S wave
- vibrate parallel to the direction.

S-Wave

- 2nd to reach
- transverse wave
- more destructive than P wave
- travels only in solid.
- perpendicular to the wave direction in the vertical plane.
- Doesn't pass through core.

Surface Wave (L-wave)

- last to report on seismograph
- more destructive.
- cause displacement of rocks.
- least speed.
- most damaging waves.
- large Amplitude.

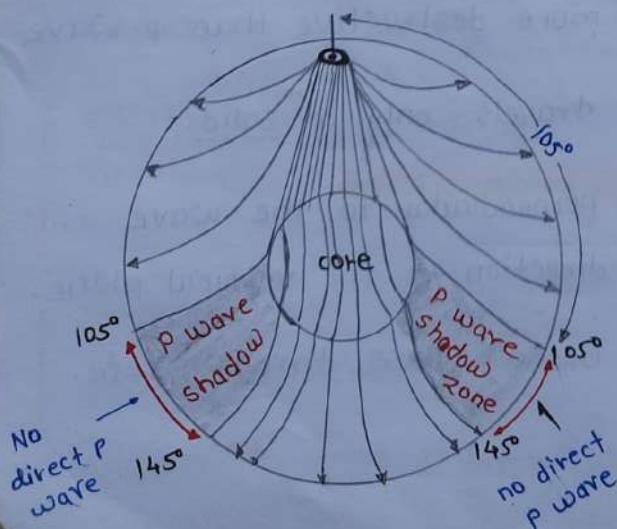
$P + S \rightarrow L \text{ wave}$

waves are not reported.

shadow zone

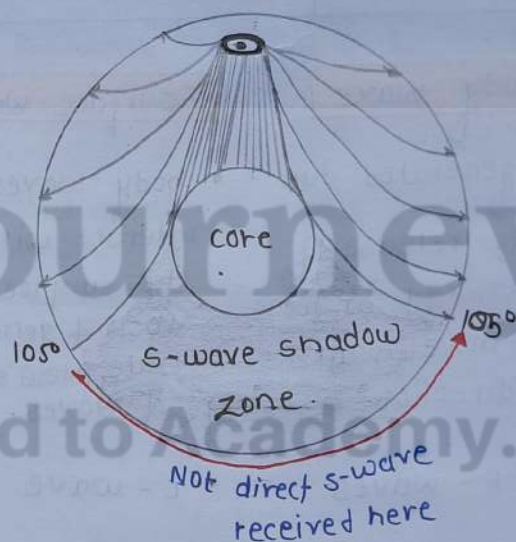
① P - wave

- Seismographs located beyond 145° from epicentre, record the arrival p-waves, x s-wave



S - wave

- entire zone beyond 105° does not receive s-waves.
 - shadow zone larger
- S-wave > P-wave



zone	Intensity
zone V	very high risk zone area liable to shaking IX
zone IV	High risk zone (intensity VIII)
zone III	Moderate risk zone (intensity VII)
zone II	Low risk zone VI + lower.

zone	state
(red) zone V v-h risk	part of Kashmir, N-E, UK, HP, A & N, Rann of Kutch
zone IV (High risk) (orange)	Indo- Gangetic basins, Delhi, J & K, MH, Patna.
zone III (moderate) (yellow)	part of Kashmir, Western Himalayas.
zone II (low risk) (blue)	MP, Tel., E-Raj., Jharkhand, Karnataka.

Zone I NO risk
(green)

Richter Scale

measure magnitude of earthquake.

magnitude : 0-9

Least - 0

Highest intensity - 9

Mercalli scale

account visible damage caused

magnitude : 1-12

Volcanoes

lower density

not much heat up

K magma

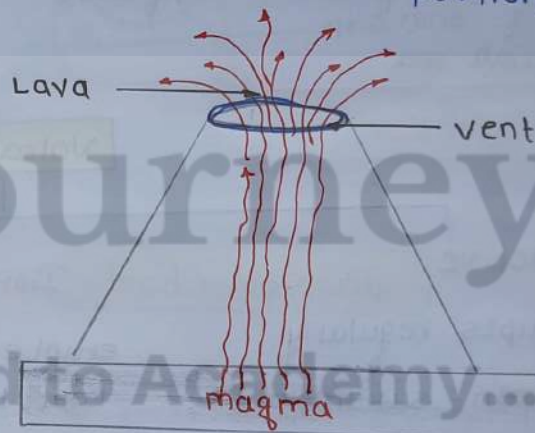
higher density

mantle > crust

asthenosphere

source of magma

upper mantle position.



⊕ Eruption from volcanic activity

* Gases

• Water Vapour (H_2O) 60-90%.

• CO_2

• NO_2

• SO_2

⊕ positive effects of Active Volcano

• Volcanic ash & dust @ very fertile.

• best for Agriculture.

• Yellowstone N.P - volcanic eruptions

Lava

Acidic Lava

- contain more silica
- viscous
- lighter in color
- creates dome structure

Ex: • Stromboli, Italy
• Puy de dome, France.

light
house of
Mediterranean sea.

Basic Lava

- contain less silica
- less viscous
- darker in color
- creates flat-top structure

Ex: • Mauna Loa, Hawaii

Volcanoes

Active

- Erupts regularly
- Japan - Sakurajima
- Hawaii Island
 - Kilauea
 - Mauna Loa
 - Mauna Kea
- Indonesia
 - Tambora
 - Merapi
 - Dukano
- Italy -
 - Stromboli
 - Mount Etna
- India
 - Barren Island

Dormant

- erupted in near past but to the potential to do
- India
 - Narcondam (Near North Andaman)
 - Indonesia
 - Krakatau ✓
 - Japan - Fujiyama
 - Italy - Vesuvius
- Tanzania - Mt. Kilimanjaro
 - Ecuador - Chimborazo
 - Argentina - Aconcagua
 - Myanmar - Popa

Extinct

Hasn't erupted in human history + not future.

Extinct

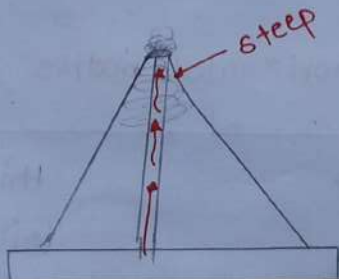
Shield Volcano

- largest volcano of earth
- made of basalt
- very fluid
- this reason volcanoes are not steep.
- low - explosivity
- If somehow water gets into vent - explosive
- develops into cinder cone
- Ex: mauna loa



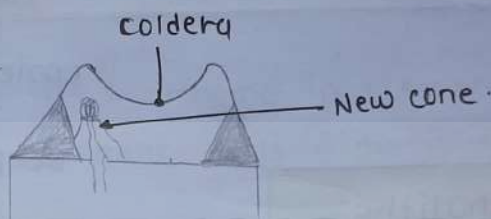
Composite Volcano

- more viscous lavas than basalt.
- cooler than shield volcano.
- Along with lava, pyroclastic material & Ashes come.
- more steep than shield vol.



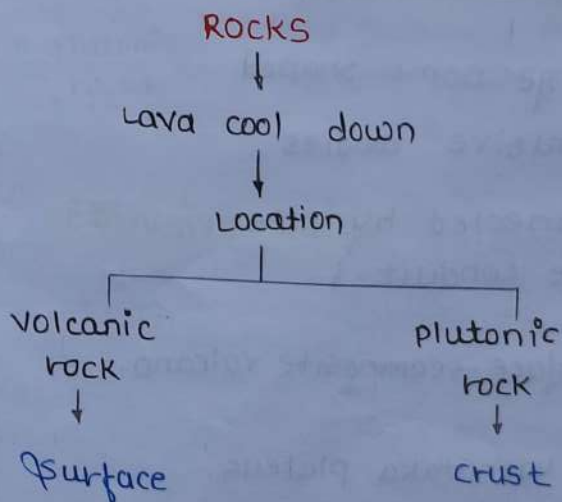
Caldera

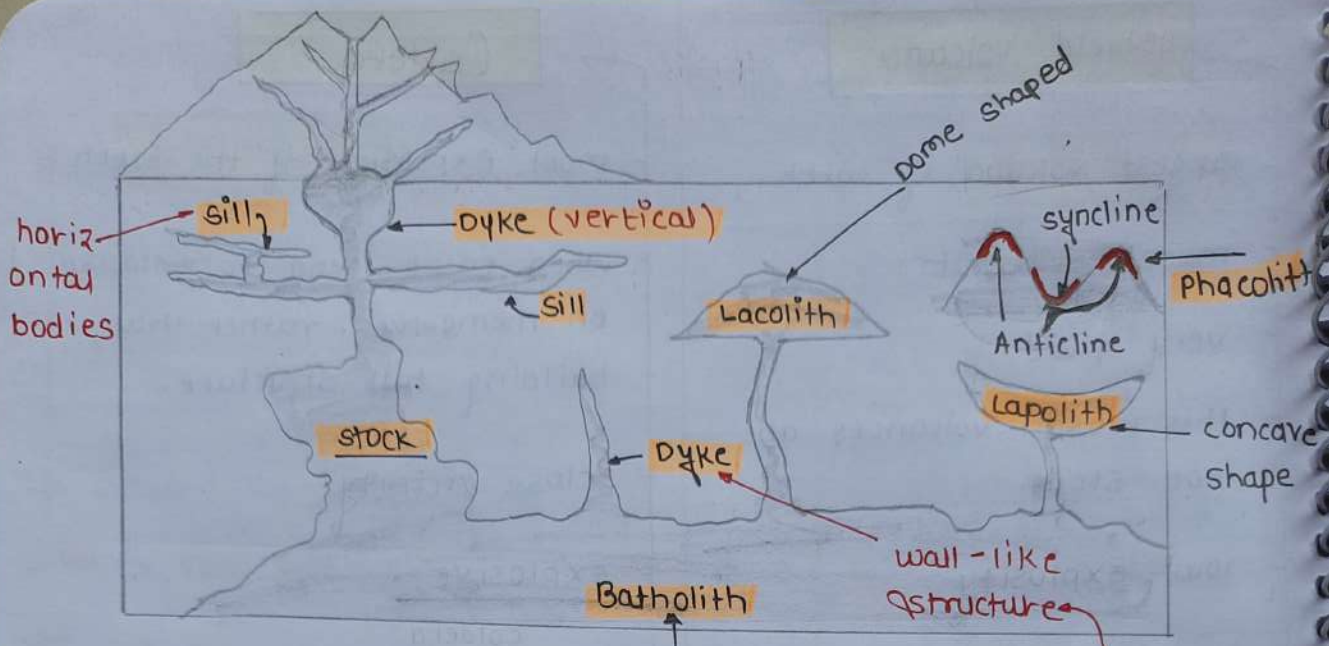
- most explosive of the earth.
- when erupt tend to collapse on themselves. rather than building tall structure.
- close vicinity
- explosive



Flood Volcano

- highly fluid lava
- flows for long distance
- Deccan traps from India.





cooled portion of magma
↓
deeper region
↓
cool down

solidifies
↓
perpendicular
to the ground

Batholiths

- magmatic material that cools in the deeper depth.
- develops - large dome
- cooled portion of magma
- granitic bodies

Lapolith

- saucer shaped
- concave to the sky body

Laccolith

- large dome shaped intrusive bodies
- connected by pipe like conduit.
- surface - composite volcano.

Ex: karnataka plateaus.

Phacolith

- Deposition on syncline
- ⊗ Anticline structure



* horizontal bodies

thinner
(sheets)

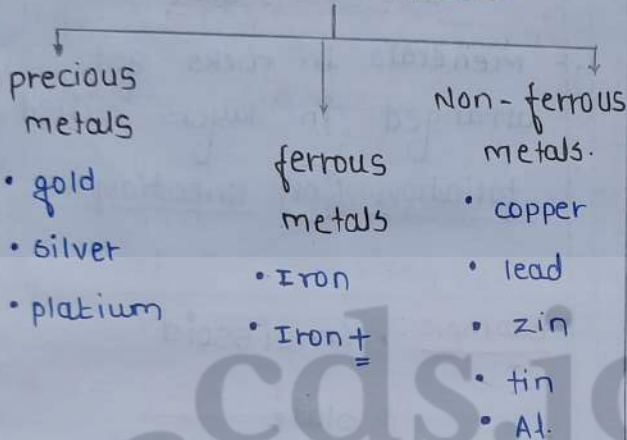
thick
(sills)

Rocks

- basic source of all minerals is the hot magma in the interior of the earth.

*

Metallic Minerals



* Most common minerals found in rocks.

- # Feldspar
- # quartz

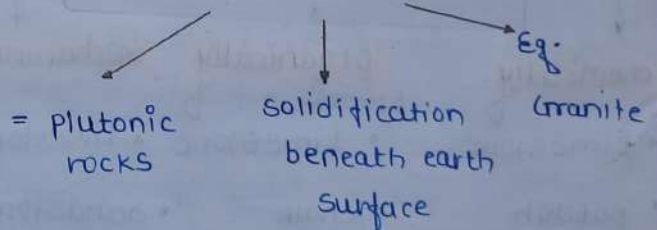
Science of Rocks - petrology

- gabbro - black
- quartzite - milky white
- granite - hard.
- soapstone - soft

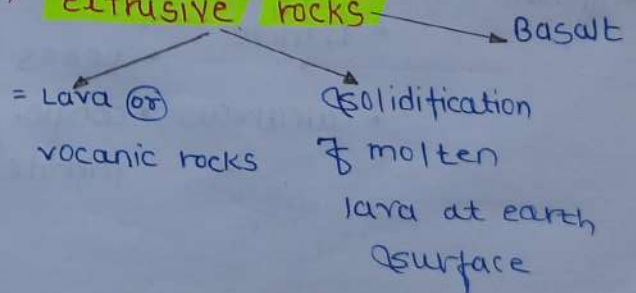
Igneous Rock

- primary rock
- most abundant (95%)
- Solidified from magma & Lava.
- oldest of all the rocks.
- When magma in its upward movement cools & turn into solid.
- Example:
 - Granite
 - gabbro
 - pegmatite
 - basalt
 - tuff
 - Volcanic breccia.

1) Intrusive rocks



2) Extrusive rocks



3) Hypabyssal Igneous rocks

Dyke or

Intermediate rocks

cool just beneath the earth's surface.

- Batholith
- Laccolith
- Placolith
- Sills
- Dykes.

Sedimentary Rocks

Detrital rocks

many layers

formed by lithification.

holds all reserve of coal, oil and natural gas.

not as rich in minerals of economic value as Igneous rocks.

chemically

- Limestone
- potash
- gypsum

organically

- Limestone
- chalk
- coal
- Lignite
- calcareous.

mechanically

- Limestone
- sandstone
- shale
- loess
- conglomerate.

Metamorphic Rocks

change of form

process of recrystallization & reorganization of minerals within the original rocks.

minerals in rocks get arranged in layers called foliation or lineation.

Example: Gneissoid

- slate
- schist
- marble
- quartzite.

two main constituents of granite

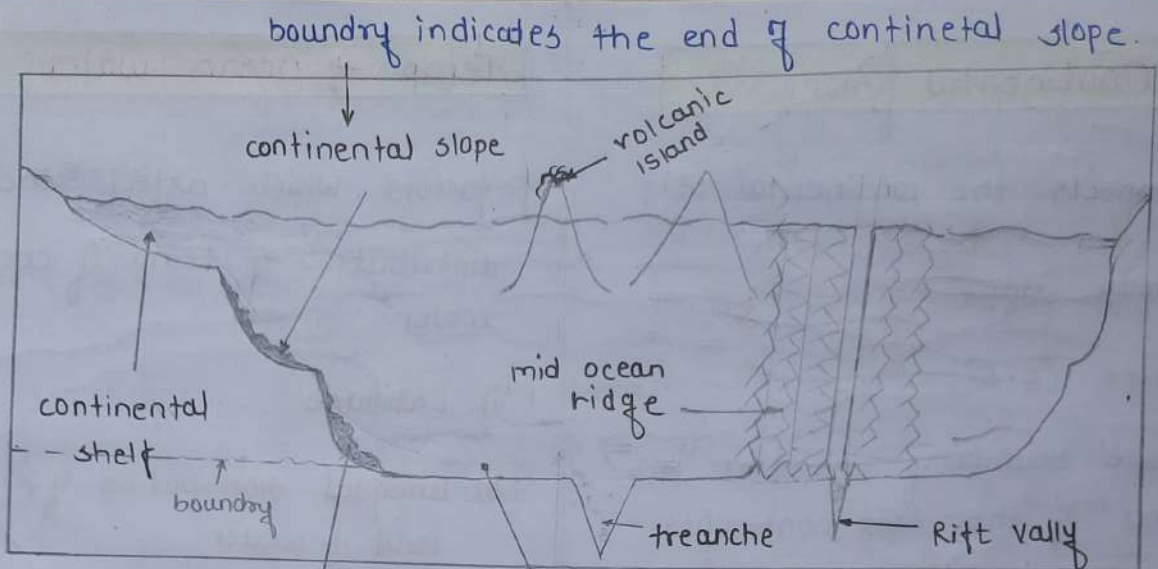
- silica
- aluminium.

lower most of innermost Intrusive igneous rocks

Batholiths

single element mineral.

- gold - graphite - copper.
- silver - sulphur



covered clay + silt

continental rise (0.5-1°)

Abyssal plain (plane area)

④ - major divisions

- ① continental shelf
- ② continental slope
- ③ deep sea plain
- ④ oceanic deeps

* minor relief features

- 1) hills
- 2) sea
- 3) mounts
- 4) guyots - plateau
- 5) trenches - convergent plate boundary
- 6) canyons
- 7) peak.
- 8) Ridge - divergent plate boundary
- ⑨ seamount - 3000-4500m tall
Ex - Hawaiian Islands.

continental shelf

- shallowest part of the ocean.
- average gradient of 1° or even less
- very steep slope
- also called shelf break
- Siberian shelf in Arctic ocean largest in the world.
- source of fossil fuels
- Good for fishing ground.

Continental slope

- connects the continental shelf
- steep slope
- slope ($2-5^\circ$)
- slope boundary indicates the end of the ~~step~~ continents.
- canyons & trenches observed this region.

Salinity

- Total content of dissolved salt in sea water.
- depends on precipitation & evaporation.
- wind (influence salinity)
- ocean current.
- Highest salinity in water bodies
 - Lake van in Turkey - 330%
 - Dead sea - 238%
 - Great salt lake - 220%
- highest salinity is recorded b/w 15° & 20° latitudes
- mediterranean sea records higher salinity due to high

Temp. of Ocean water

* factors which affect the distribution of temp. of ocean water.

- i) Latitude
- ii) unequal distribution of land & water
- iii) prevailing wind
- iv) Ocean currents.

Ocean Currents

- primary factors influence of oceanic currents.

- ① Coriolis force - Because of rotation of earth
- ② Gravity
- ③ wind
- ④ Heating by solar energy.

- Secondary factors.

- ① salinity
- ② temp.

- Black sea - very low. (freshwater)

- Indian ocean - 35% .
evaporation

Coriolis force.

Because of rotation of earth.

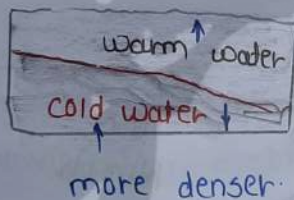
zero @ equator - cyclone (X)

N.H - Right } $\frac{\text{बुझाता है}}{2}$
S.H - Left

* which is denser

① cold water - sink

② warm water



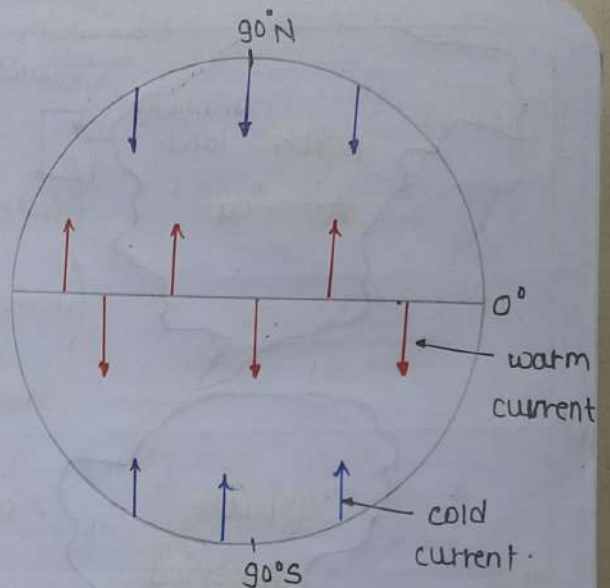
* factor which influence ocean currents

① Rotation

~~② Revolution~~

② Air pressure

③ Density of ocean.



* warm current

- $0^\circ \rightarrow 90^\circ$ (movement)

- equator to pole

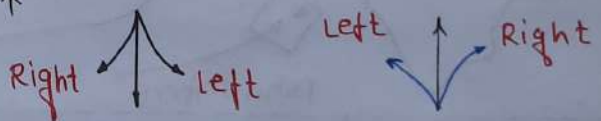
* cold current

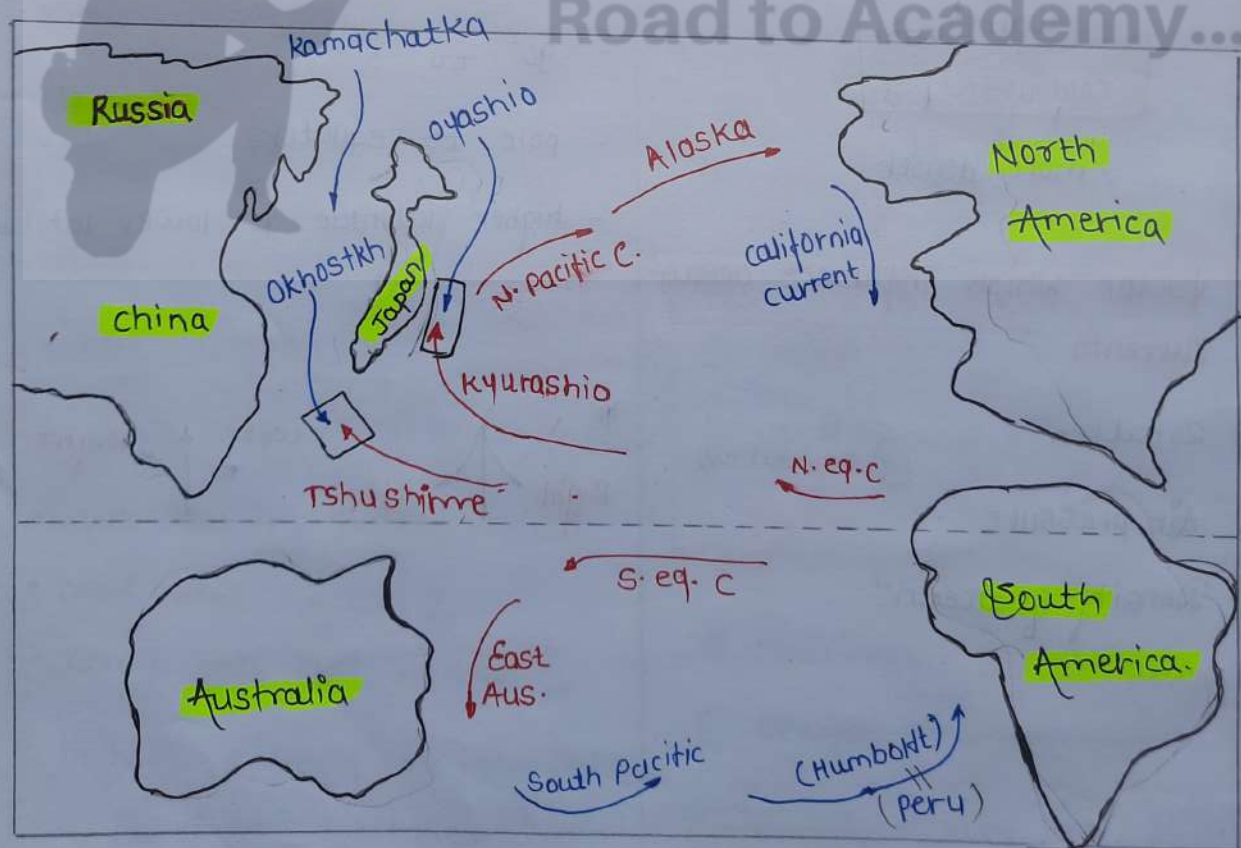
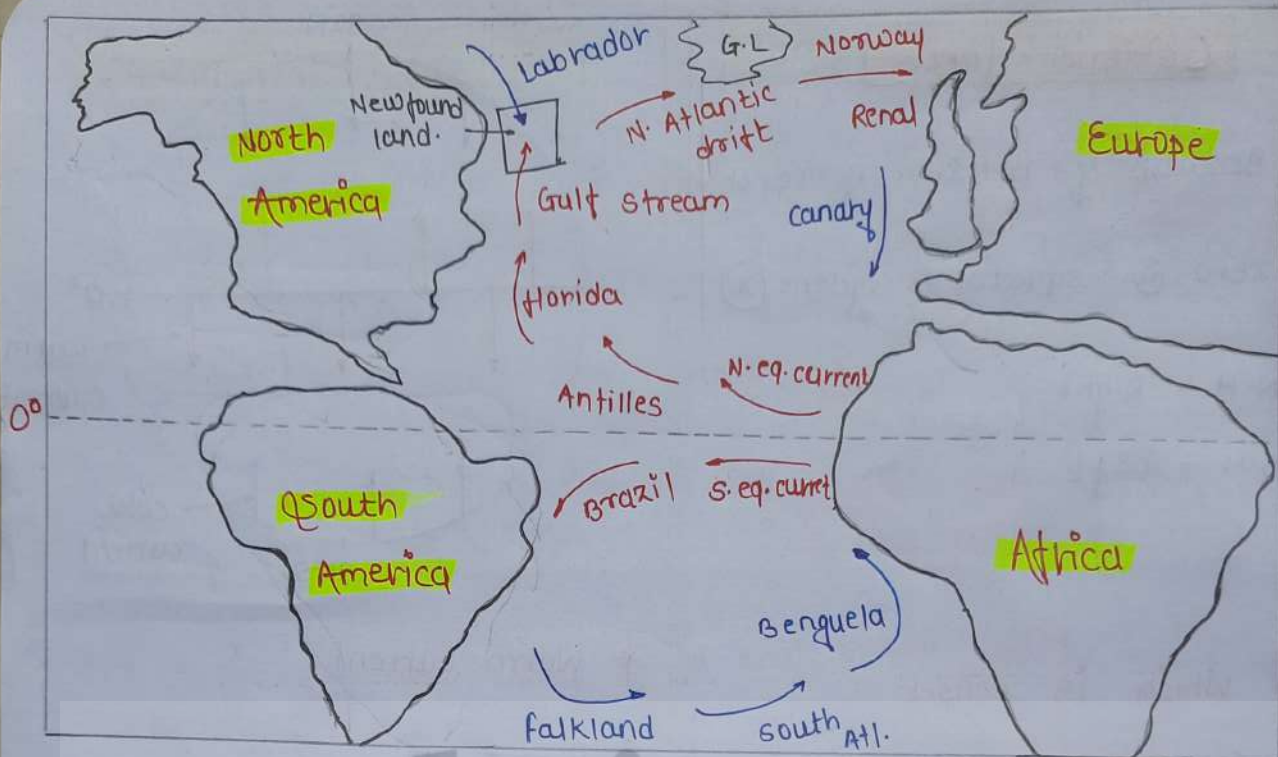
- 90° to 0°

- pole to equator

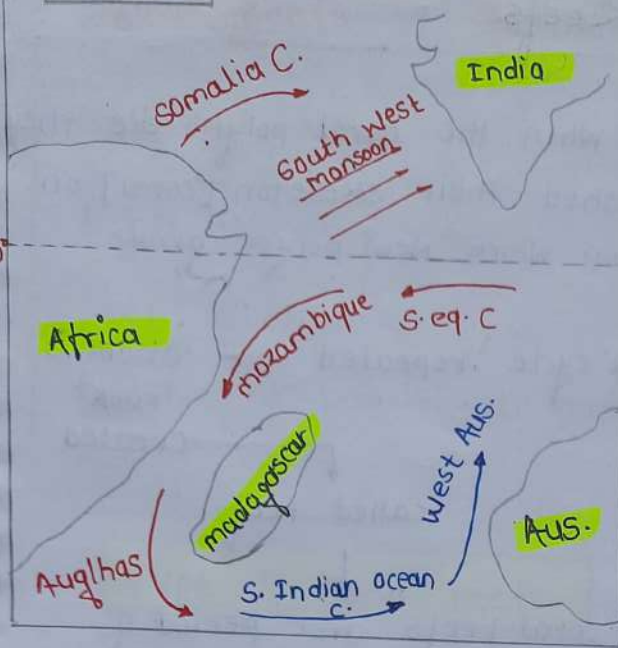
- higher latitude to lower latitude

*

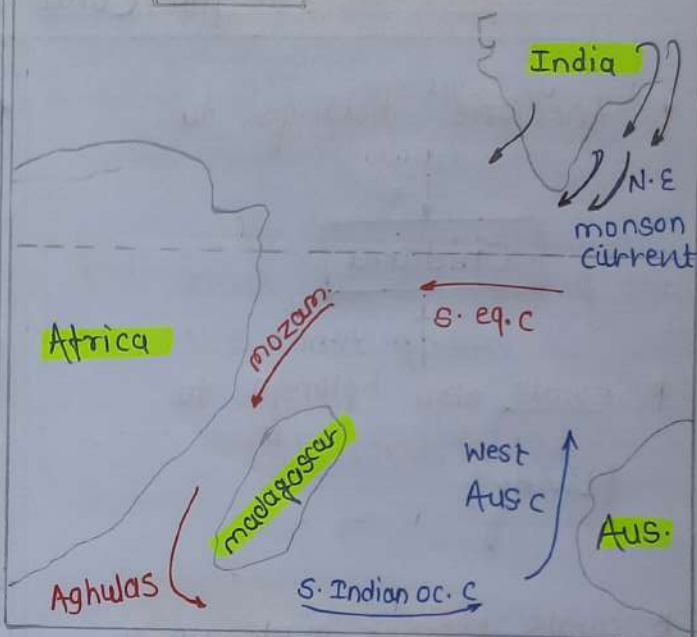




Summer



Winter



* Meeting point

↓
warm + cold

- ① foggy environment
- ② navigation problem
- ③ good fishing ground.

↓
warm + cold

↓
help to replenish the O_2 &
favour the growth of
phyto-planktons

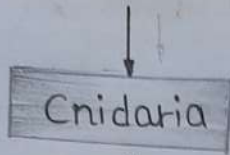
↓
primary food for fesh.

* cold current form desert coastal

- Atacama desert (chile)
- Crobi desert (m+c)
- Ladakh

The Coral Reefs

* Anemone belongs to



* corals also belongs to
(Living)

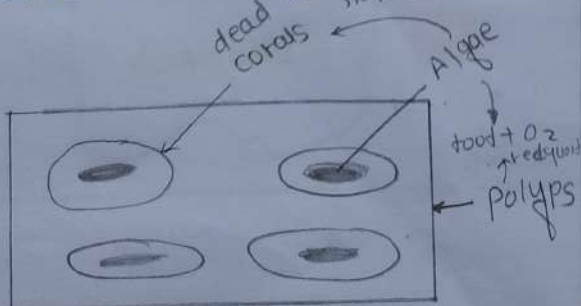
* corals reefs - made up of thousands of tiny animals called "polyps"

* **Neritic zone**

- maximum corals found in Neritic zone
- 200m deep
- allowing sunlight to reach the entire seafloor over the continental shelf.

- part of the ocean's photic zone

- coast से थोड़ा दूर.



* When the coral polyps die they shed their skeleton [coral] on which new polyps grow.

* cycle repeated → shallow rock created
called reef.

coral reefs over period of time

↓
coral Islands

↓
Ex: Lakshadweep.

* polyps have symbiotic relationship with zooxanthellae (algae)

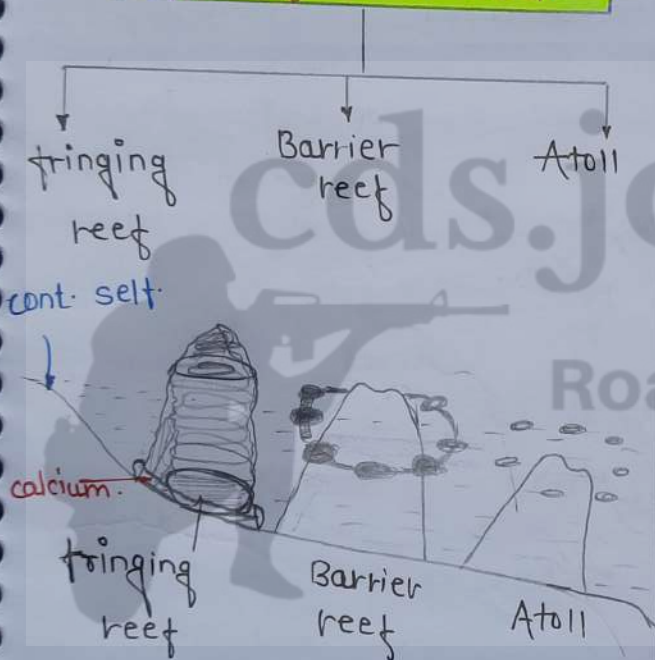
Ideal conditions for coral growth

- shallow water
- Semi-hard or hard surface
- Saline water
- clear water
- warm water
- Rich supply of nutrients
- Abundant plankton
- NOT fresh water.

Coral Reefs in India

- Gulf of Mannar
- A & N Islands - maximum
- Gulf of Kutch
- Lakshadweep Islands - all made.

Type of Coral reefs



fringing reef

- most commonly found reefs in the world
- Example: • South Florida reefs
- Sakau Island in New Hebrides.

Barrier Reefs

- found offshore on the continental shelf.
- Ex: Great Barrier reef of the N-E coast of Aus.
↓
largest coral reef.

Atolls reefs

- ring shaped coral reef island
- Ex: • Fiji Atolls
- Funafuti Atolls of Ellice
- Suva Diva in Maldives
- Chagos Island.

Causes of Coral Bleaching

- High temp.
- Extreme low tides
- ocean acidification
- Diseases
- ocean pollution

Threats to Coral reefs

- Global warming
- coral bleaching
- marine pollution.
- ocean acidification
- ozone depletion
- invasive alien species
- over exploitation
- poorly managed tourism

Coral Health Important

- marine diversity
- Tourism
- fishing
- coastline protection
- medicinal properties

* coral reef island

- Great Barrier Reef, Aus.
- Rainbow reef, Fiji
- Pwarray Island, India. (A+M)

* largest barrier reef system in the world

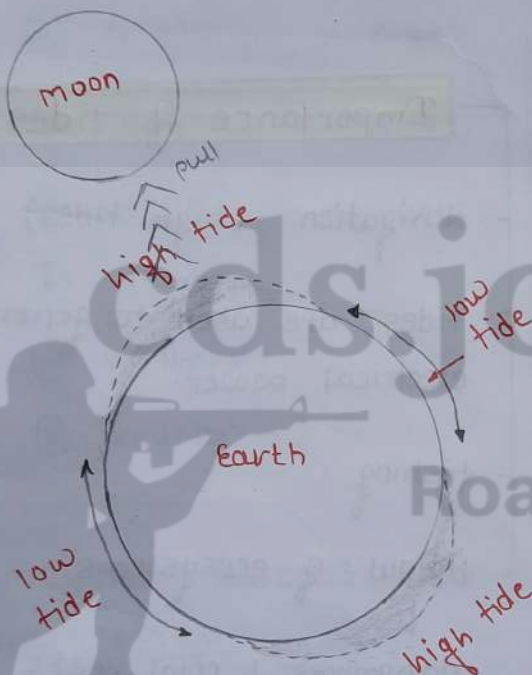
↓
East / N-E Australian coast.

Tides

→ upward & downward movement

* responsible for creating tides

- gravitational pull of moon (main) (great extent)
- centrifugal force
- gravitational pull of sun. (lesser extent)



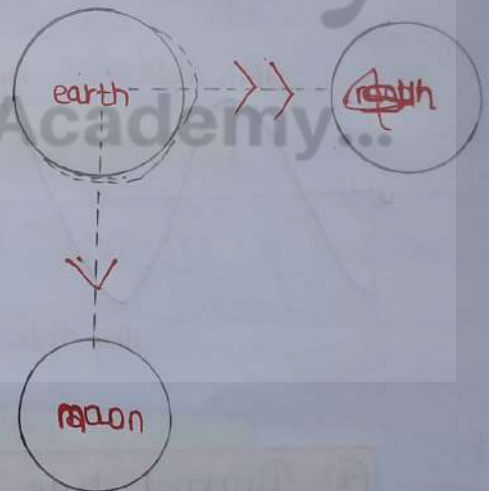
Neap tides - (twice month)
(1st & last quarter of month)

- seven day interval b/w Spring tide & Neap tides.

- Sun & moon right angle to each other

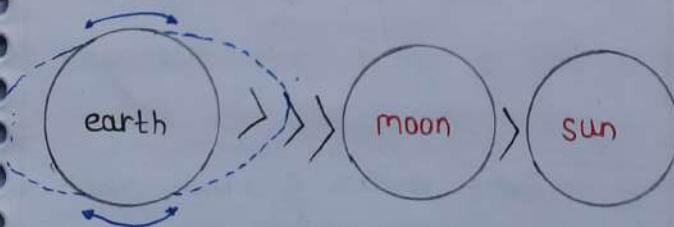
↓
force contract one another.

- attraction → moon > sun (twice)



* **Spring tide** - highest tide

occur twice a month.



ebb - time b/w the high tide and the low tide, when water level is falling.

Bay of Fundy - Highest tidal range (Canada)

① full moon period

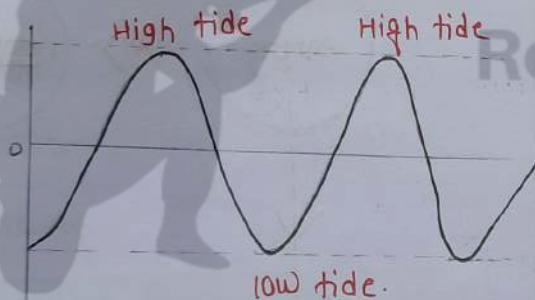
② New moon period.

Tides based on frequency

- Semi-diurnal tide
- Diurnal tide
- mixed Tide.

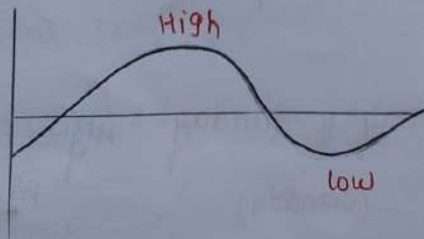
① semi-diurnal tide

- two high tides & two low tides each day (same height)
- interval (high & low) - 12 hr, 25 min
- most widespread in Indian ocean.



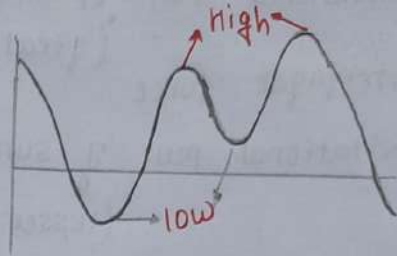
② Diurnal tide

- one high & one low tide (same height)



③ mixed tide

- variations in height
- both ① & ② oscillations.



Importance of tides

- navigation (high tides)
- tides are used to generate electrical power.
- fishing
- Helpw of ecosystems
↓
mangroves & coral reefs grow & sustain.

Climatology

Atmosphere

* gases in Atmosphere

- ① Nitrogen - 78%
- ② oxygen - 21%
- ③ Argon - 0.93%
- ④ CO₂ - 0.03%
- ⑤ Neon - 0.0018%
- ⑥ Helium
- ⑦ methane
- ⑧ krypton

* Atmosphere extends upto about 1600 km.

Role of earth's Atmosphere

protect from harmful gases like UV ray.

Regulate the temp. of earth

take care of extra terrestrial objects like (meteors).

Radiation :

- doesn't need any medium for heat transfer.

↓
without contact

- Ex: Sun energy $\Rightarrow$ earth

Insolation

- amount of Sun's energy received in the form of radiation.

Convection

- Actual transfer of matter.

Albedo

- proportion of sunlight that the surface can reflect back into space.

Black

↓
all light absorb and reflect none.

White

↓
absorb none reflects everything.

Albedo of
White (light)
object

Albedo of
Black object.

- fresh snow
- thick cloud
- water
- old snow
- concrete

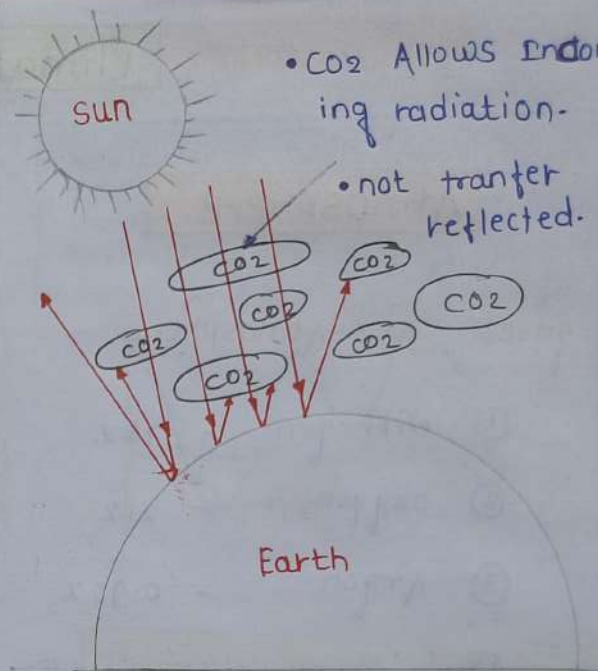
- Desert
- soil
- Grass

maximum Albedo in planets

Venus - made from
sulphuric Acid
↓
very reflective

maximum Albedo

- ① Tundra
- ② Taiga
- ③ Tropical dec.
- ④ tropical evergreen.



temp ↑

GHG - Kyoto protocol

- CO₂ - 84% • CFC
- methane - 9%
- Nitrous oxide (N₂O) - 6%
- fluorinated gas - 3%
- water vapour

Ozone (O₃)

- Stratosphere
- Earth's sunscreen
- filter & absorbs the UV rays
- relative thickness is measured in Dobson units.

Aurora's observed in lower parts of this layer

Exosphere (temp ↑)
density of air very less

640 km

- ISS
- satellites

Inosphere (Temp ↑)

Thermosphere present

- radio wave (transmission)

80 km

(vacuum)

Mesosphere (Temp ↓)

- coldest layer
- meteorite Burn

Stratopause constant

50 km

Stratosphere (temp ↑)

- ozone
- Jet Fly

Tropopause constant

8 km

- weather changes
- clouds

Troposphere (Temp ↓)

called changing sphere

18 km

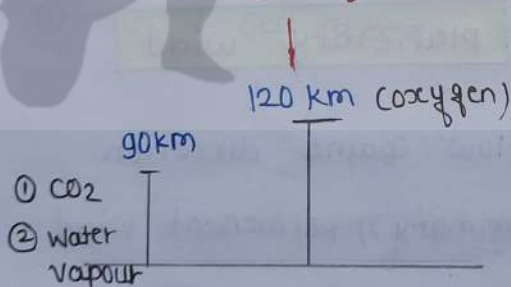
temp ↓ every 1 km - 6°C

165 m of height

pole

equator

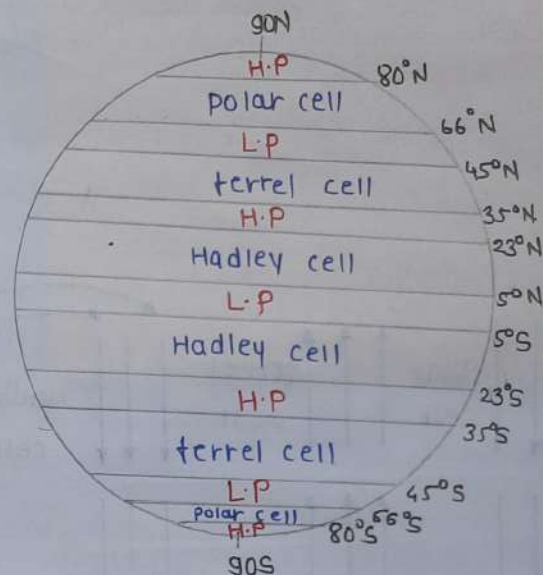
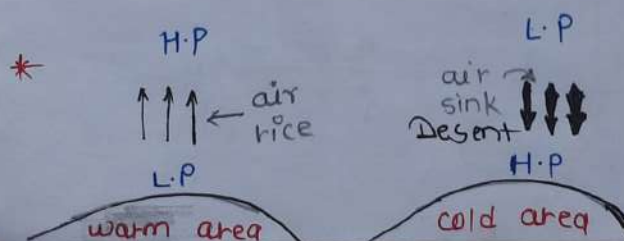
* oxygen is negligible quantity at the height of atmosphere



* gases → less - less pressure

↓

more - more pressure

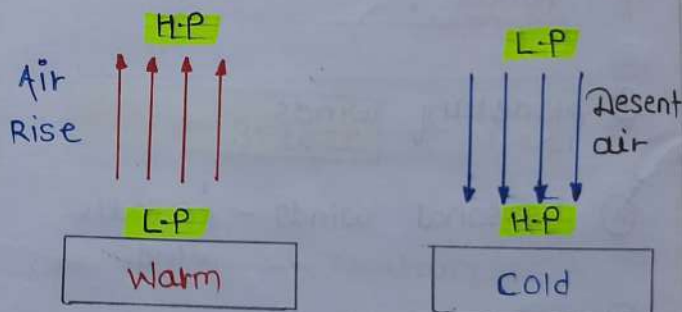


* Atmospheric pressure unit

mb + pascals

* Wind flows

H.P → L.P



* winds

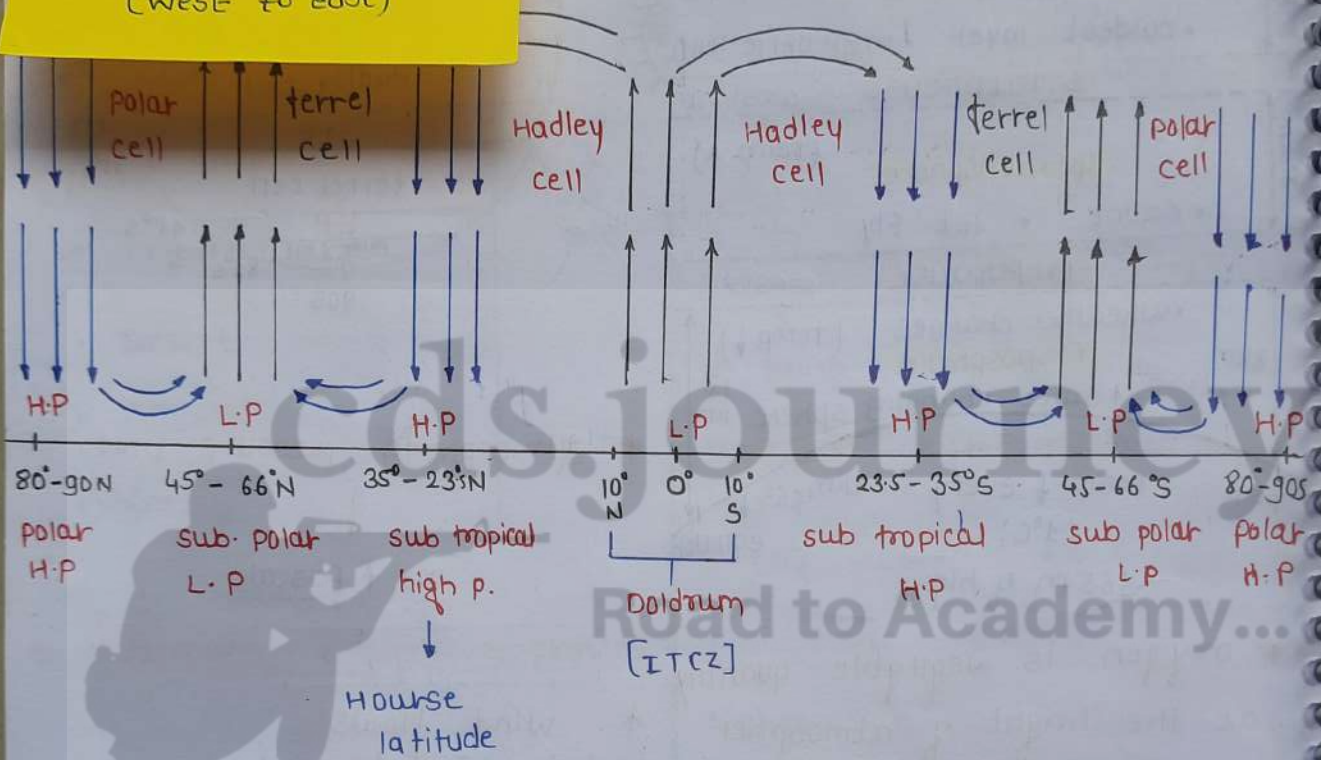
- Roaring - 40's
- Furious - 50's
- Shrinking - 60's
- screaming 70's

Jet streams

High moving upper tropospheric
Westerlies. (speed - 300km/h)

upper Air circulation

High speed tropospheric/
geostrophic winds.
(West to East)



* Winds :

- ① planetary winds
- ② Seasonal winds - periodic wind.
- ③ Local winds

planetary wind

- flow same direction
- primary / permanent wind
- ① trade winds
- ② westerlies
- ③ polar easterlies

* frictional force

Affect the speed
of wind

Coriolis force

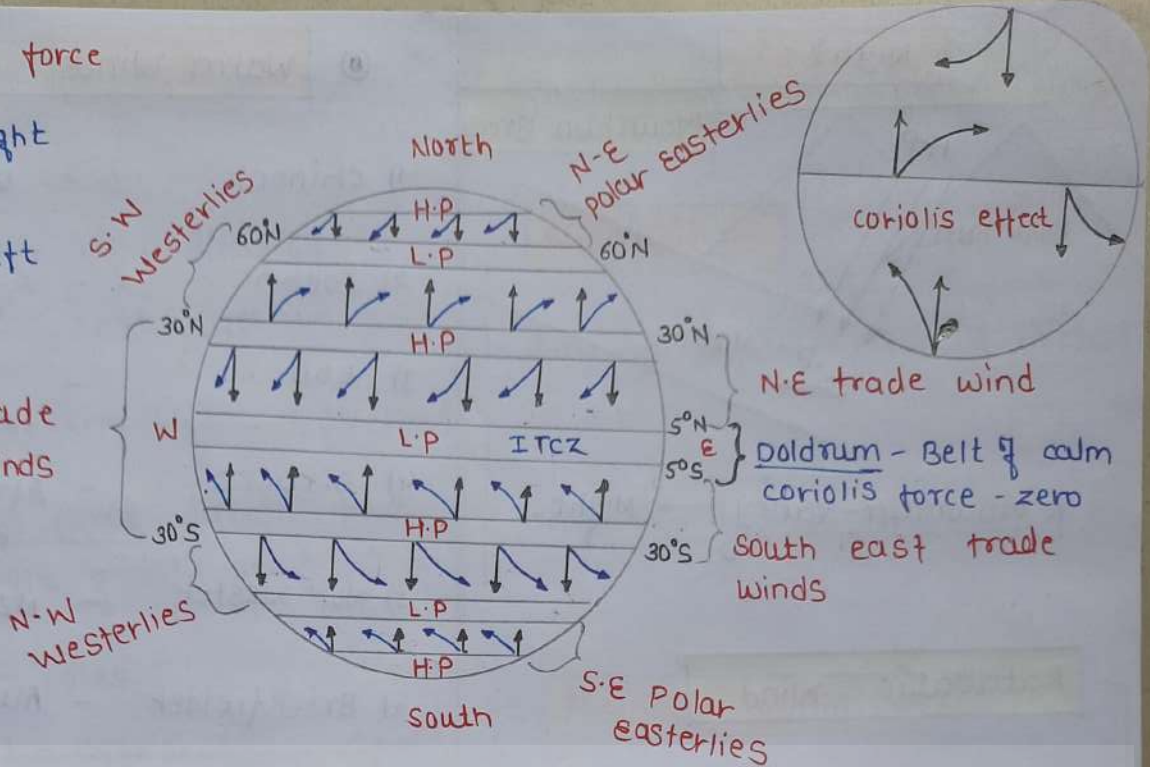
affect the
direction of
wind

coriolis force

NH - Right

SH - Left

Tropical Zone Trade winds



Seasonal winds

- periodic winds
- can change daily direction (diurnal)
- 3 types
- ① Land & sea Breeze



land - cool down faster
sea water - slow heat
slow cool

Land sea Breeze

L.P. sea water H.P. land

heat faster also
cool faster

Land - heat up faster
sea water - slow heat up

Day

Sea land Breeze

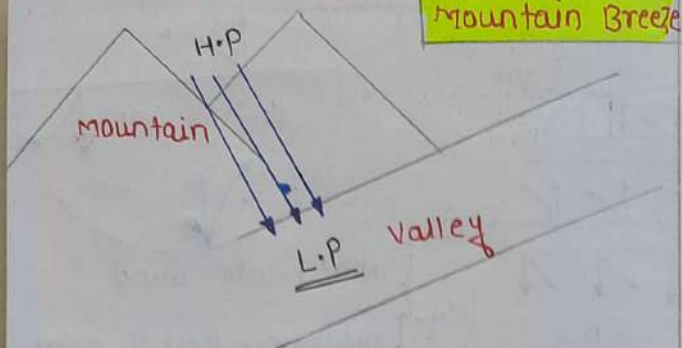
H.P. sea water
Land coastal region
Sea to land

Valley Breeze

Day
Valley -> mountain
L.P. warm area
H.P. Valley cool area

Night

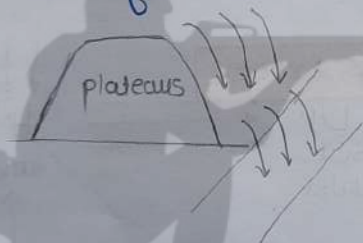
Mountain Breeze



* Mountain-valley → Night

Katabatic Wind

The cool air, of the high plateaus and ice fields draining into the valley is called.



Local winds

① Cold Winds

- 1) Pamero - Argentina
- 2) Gregale - mediter. sea
- 3) Bora - Adriatic coast (Hungary, N. Italy)
- 4) Mistral - France
- 5) ~~Harmattan~~ ^{purga} - Africa Russia
- 6) Lavanter - France & Spain
- 7) Nor'easter - Canada.

② Warm Winds

- 1) Chinook - USA
- 2) Foehn - Switzer, Alps
- 3) Loo -
- 4) Sirocco - Africa & south Europe
- 5) Nor' Wester - NZ
- 6) Brickfielder - Aus
- 7) Kham sin - Egypt
- 8) Harmattan - West Africa
- 9) Karaburun - central Asia
- 10) Elphanta - malabar coast
↓
moist wind in monsoon

③ The horizontal Winds near the earth's surface are affected

- ① pressure gradient force
- ② frictional force
- ③ coriolis force

cold	Warm
• cool & dry	• Warm & dry
• chilling effect	• warm & moist.
• ↓ temp.	• ↑ temp.
• Not good for vegeta.	• melts snow
	• good - vegetation in cold

Coriolis force

- deflect wind :
 - N.H - right direction
 - S.H - Left direction
- Wind velocity high - deflection $\uparrow$
- Coriolis force directly prop. to angle of latitudes.
- max - poles
- absent - equator - tropical cyclone not formed near equator.
- P. gradient force Perpendicular to isobar.

Inversion of Temp.

- Normally : Troposphere
 $\uparrow 1\text{km} \rightarrow 6^\circ\text{C} \downarrow$
 - Temp. $\downarrow$ with increasing Altitude
- In Inversion - Temp. rise (increase) with Altitude.
 $\downarrow$ opposite behavior.
- Ideal condition
 - ① Long winter Night
 - ② clear sky
 - ③ slow movement of air
 - ④ snow covered surface. (pole)

$\downarrow$
very common at higher latitude
- Impact - less rainfall.

Water in the Atmosphere

Humidity

- Amount of water vapour present in Air
- measurement - Hygrometer

Absolute humidity

- Actual Amount of Water Vapour present in the Atmosphere.

Relative humidity

- percentage of moisture present in the atmosphere as compared to its full capacity at given temp.

Dew point

- temp at which water vapour begins to change into
liquid (or) solid

Freezing point

temp @ which liquid start turn solid.

0°C

Evaporation

Water - liquid $\Rightarrow$ Water Vapour.

factors:

evaporation increases with.

- Increase of temp.
- Increase of Surface Area
- Increase in wind speed
- decrease in humidity

↑ movement of air

actual Amount of water vapour present

- amount of water

Condensation

transformation of water vapour into water / solid.

more Evap. - more conden

- sublimation - water vapour directly condenses into solid.

* condensation take place

① When temp. of air is reduced to dew point.

② with volume ^{remains} constant

③ both volume + temp. reduced

④ moisture added to air through evaporation.

⑤ decrease air temp.

⑥ freezing point - lower / higher

forms of Condensation

Dew point is Above freezing point (0°C)

Dew point is below freezing point (0°C)

- Dew
- fog
- clouds
- mist
- Haze
- smog

- Frost
- snow
- hailstones
- cirrus clouds

solid

Dew

- When moisture deposited in the form of water droplets on • stone • plant leaves • grass

(dew point)
condensation above freezing point

Ideal conditions

- ① clear sky
- ② calm air
- ③ high relative humidity
- ④ cold & long nights.

clouds prevent the escape of heat from the ground.

Frost

- condensation take place below freezing point (0°C).
- temp. less than 0°C .
- excess moisture is deposited in form of minute ice crystals. Instead of water droplet.
- Ideal condition - Same as dew except - temp must be $\text{at } (or) \text{ below } (0^{\circ}\text{C})$

fog

Temp. $0^{\circ}\text{C} +$

- very near to the ground.
- visibility $\downarrow$ - poor to zero (mist)
- fog are mini cloud
- fogs are drier than mist.

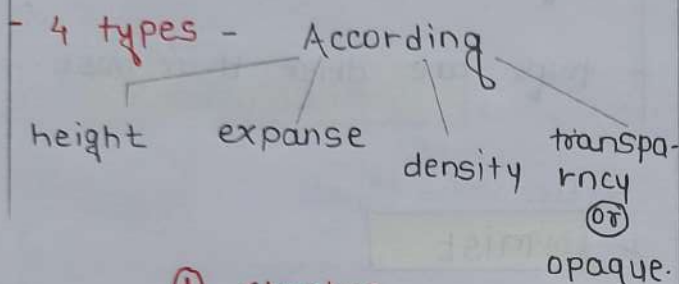
mist

- fog & mist same
 $\downarrow$
only difference
 $\downarrow$
mist contain more moisture than fog.
- frequent over mountains
- visibility $\uparrow$ - more as compared to fog.
- smog = smoke + fog

Haze	smog
- dust, smoke	- smoke + fog
- No condensation	- condensation ✓
	- fog - w.v $\rightarrow$ liquid + smoke

cloud

water droplets (or) tiny crystal of ice formed by condensation of water vapour.



① stratus

② nimbus

③ cumulus

④ cirrus

stratus

- lower layer (altitudes)
- horizontal layering
- colour - dark gray / blue to nearly white
- covering large portion of sky
- generally formed due to loss of heat (or) mixing of air masses with different temp.

Nimbus

- colour - black (or) dark gray
- form - middle levels (or) very near to surface
- extremely dense
- opaque to the rays of sun.
- shapeless masses of thick vapour.
- rain (water cloud)
- nimbostratus - continuous rain

Cumulus

- look - cotton wool
- formed (height) : (4-7 km)
- flat base
- scattered here & there

cirrus

- form - high altitudes (8-10¹² km)
- thin & detached
- ice crystals (solid)
- color - Always white
- feathery appearance

* Ideal order - SNCC

④ **cirrus** - solid - ice crystal
high altitude

③ **cumulus**

② **Nimbus**

① **stratus**

high clouds

- ① cirrus
- ② cirrostratus
- ③ cirrocumulus

middle clouds

- ① altostratus
- ② altocumulus

lower clouds

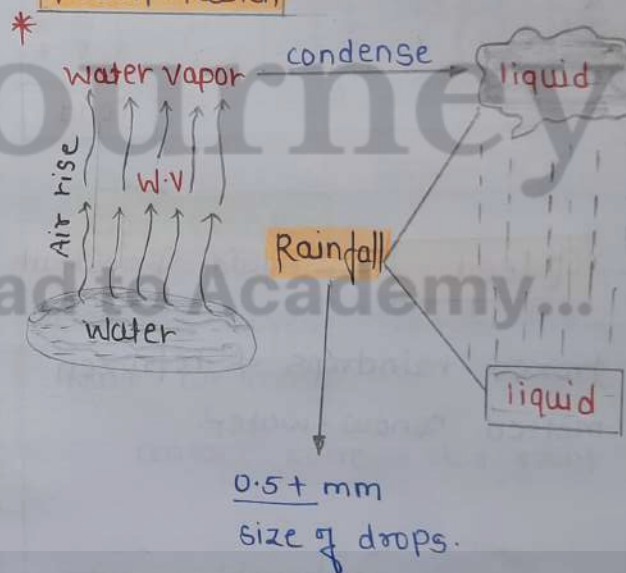
- ① stratocumulus long dur.
- ② nimbostratus - max. rainfall

◎ cumulonimbus

↓
thunderstorm

sun's halo	Rainbow
• refraction	• refraction
• reflection	• reflection
• 22° bend towards observer.	• dispersion ✓

precipitation



Drizzle - light rainfall

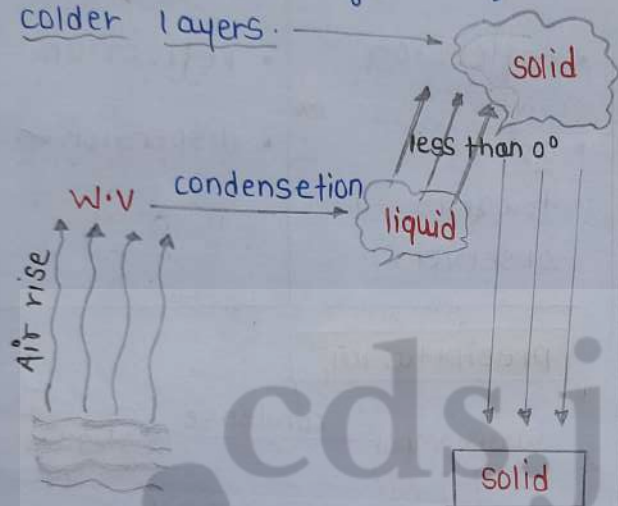
↓
drops size - less than 0.5 mm

Virage

↓
raindrops evaporate before reaching the earth.

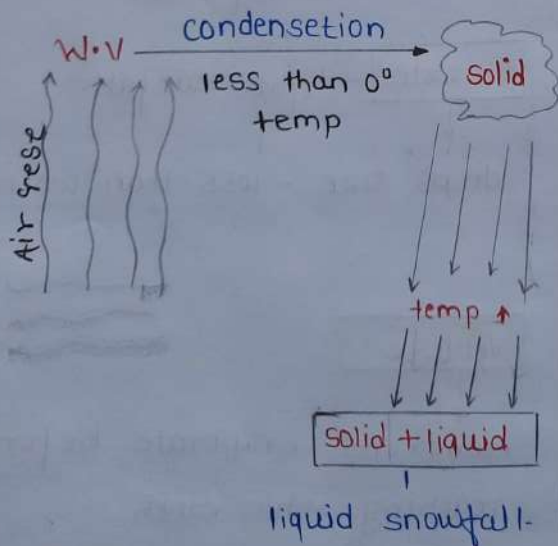
hailstones

- drops of rain released by cloud in solid form (pieces of ice).
- rainwater passing through the colder layers.



Sleet → liquid snowfall

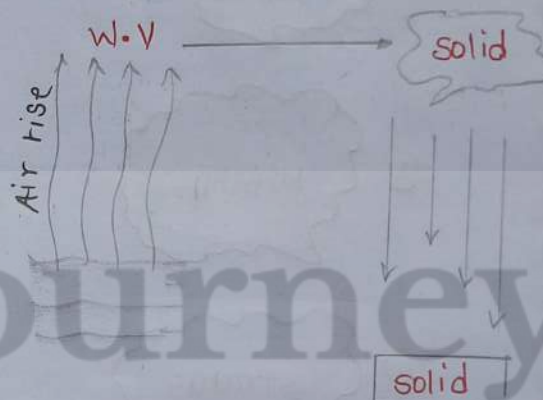
- frozen raindrops & refrozen melted snow-water.



Snowfall

- temp. - lower than 0°C
- moisture released in the form of hexagonal crystal

form flakes of snow.

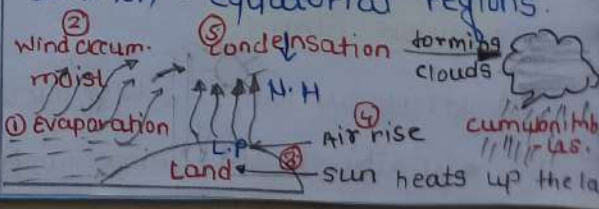


Types of Rainfall

Convectional Rain

Heavy rainfall with lightning & thunder.

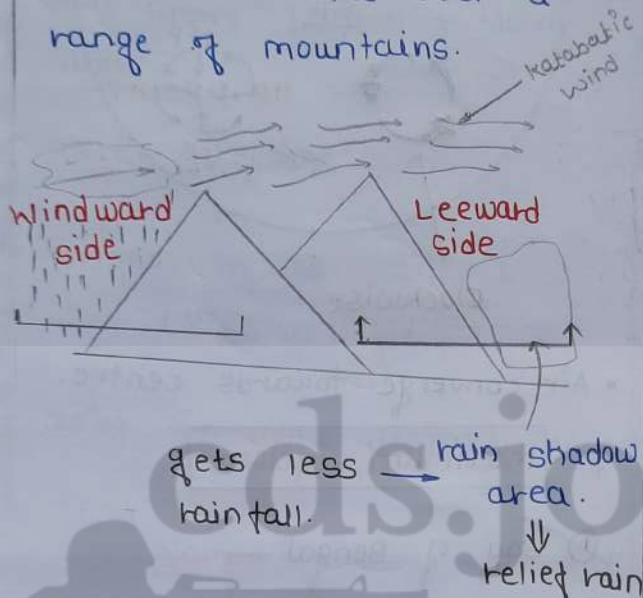
- does not last long.
- Summer (or) hotter part of the day.
- common - equatorial regions.



Orographic Rain

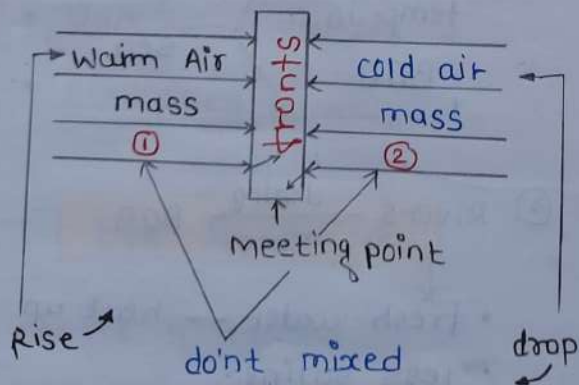
Windward slopes receive greater rainfall.

moist air moves over a range of mountains.



Fronts

two different air masses meet → boundary zone b/w them.

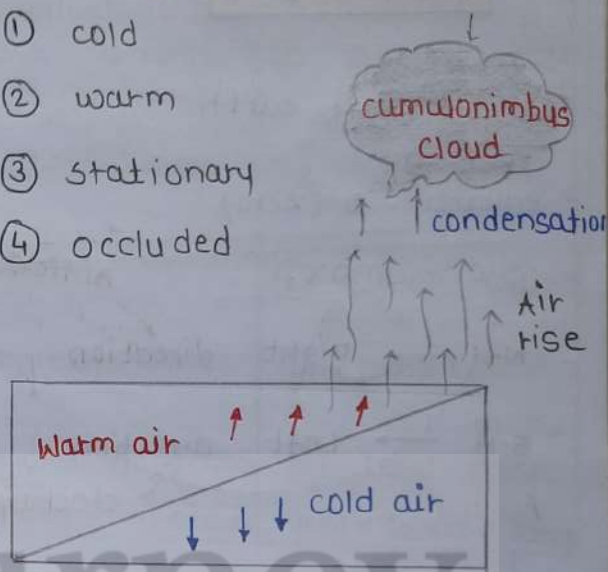


→ middle latitude
↑
occur in

process of formation front
frontogenesis.

four types

- ① cold
- ② warm
- ③ stationary
- ④ occluded



cold front

cold air moves towards the warm air mass.

↓
contact zone - cold front

Warm front

Warm air mass moves towards cold air mass

↓
contact zone - warm front

Occluded front

Air mass is fully lifted above the land surface.



cyclone

coriolis force

Rotation of earth

equator - 0 (zero)

pole - max.

N.H → Right direction

S.H → Left direction

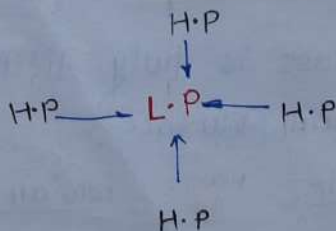
Anticlockwise

clockwise

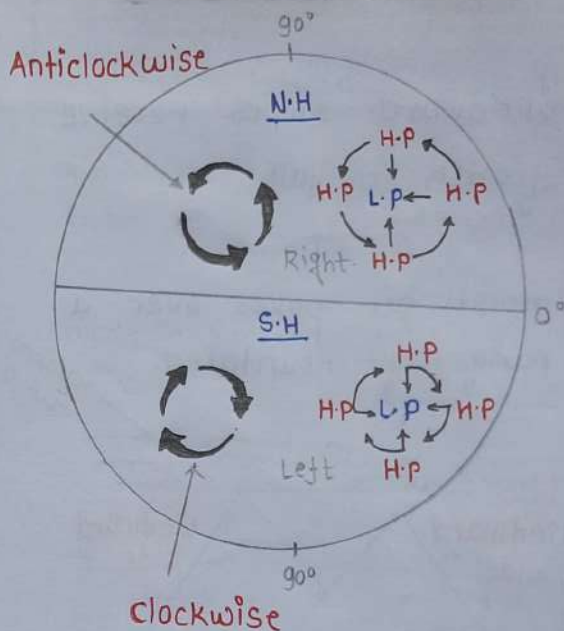
cyclones

Ideal condition for cyclones

- ① presence of coriolis force
- ② Low pressure zone
- ③ large sea surface area
- ④ temp. $27^{\circ}\text{C} +$
- ⑤ upper level divergence above sea level
- ⑥ small variation of in wind speed vertical



• develop in area of closed isobars



• Air converge towards centre.

* maximum cyclone

① Bay of Bengal

② Arabian sea

causes

① surface temp. in BOB > Arabian sea

② Rivers → draining more → BOB

- fresh water → heat up
- less saline
- strong low pressure

* salinity

Arabian sea > Bay of Bengal

Tropical cyclones

- violent storms
- originate - over oceans in tropical areas.
- very heavy rainfall + storm surges.
- more dangerous
- formed in low latitudes (8 to 20 degree)
- 20°Nt → more destructive.

Names

- Indian ocean - cyclone
- Atlantic ocean - hurricanes
- West pacific - typhoon
- Western Aus. - Willy-Willies
- USA - tornado
- Phillipines - waigu.

Eye of cyclone

- centre of cyclone
- No rain
- calm weather & clear sky
- extremely low pressure area.
- diameter of eye - 10-30 km. (roughly)

Tropical

- ① formed in low latitude - 8-20°
elliptical (shape)
- ② East to west
- ③ 100-200 km/h
- ④ ocean (summer)
- ⑤ unpredictable
- ⑥ more dangerous
speed - on ocean

temperate

- ① formed in middle latitude - 30-50°
"V" (shape)
- ② Towards east (W to E)
- ③ 30-50 km/h
(winter)
- ④ ~~ocean~~ land
- ⑤ predictable
- ⑥ Less dangerous
more (speed) land

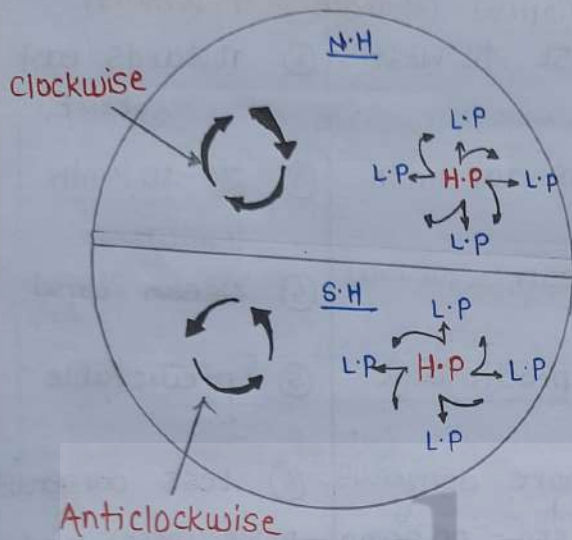
Tropical

- ① X not present
- ② originate - land + sea
- ③ ~~affects much larger area~~ ↓
- ④ wind velocity much higher ↓
- ⑤ ~~less~~ destructive more
- ⑥ move - east
to west

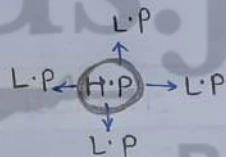
extra-tropical

- ① clear frontal system
- ② originate - seas
- ③ affects much larger area
- ④ ~~wind~~ velocity ↓ much higher
- ⑥ less destructive
- ⑥ move - West
to east

Anticyclone



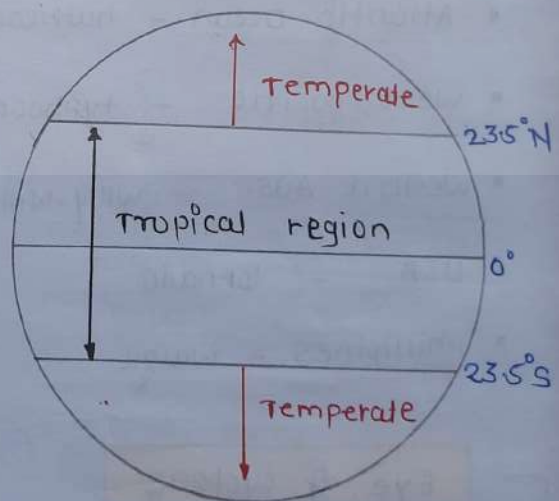
- H.P at center
- fair weather
- calm weather
- Wind blows in circular motion
- N.H - clockwise
- S.H - Anticlockwise
- Anticyclone are high pressure system.
- Air in centre of system must be subsiding.
- air diverge from centre



cyclone

Anticyclone

- | | |
|---|---|
| ① converging wind | ① Diverging wind |
| ② L.P at centre | ② H.P at centre |
| ③ | ③ |
| ④ N.H - Anti clockwise
S.H - clockwise | ④ (subsiding)
N.H - clockwise
S.H - Anticlockwise |



LANDFORM

Earth Movement

Endogenic force (internal)

Diastrophism

deformation of earth crust

- Folding
- Faulting

process - move, elevate (or) build up portions of earth crust.

through faulting & fracturing of crust.

- 1) orogeny - mountain building
- 2) epeirogeny - continental building
- 3) earthquakes
- 4) plate tectonics

All these causes
[PVT]

- p - pressure
- v - Volume
- T - Temp.

sudden mov.

- volcano
- Earthquake

Exogenic force (external)

mass wasting

deposition

Weathering

Disintegration & chemical decomposition of

- rocks
- soil

major groups of weathering

- chemical
- phy / mechanical
- biological
- oxidation
- hydration
- carbonation
- Heat.

Erosion

- Wind
- running water
- glaciers
- Ground w.
- River

Denudation

- All exogenic geomorphic proc.

weathering

⊕

mass wasting / mov.

⊕

errasion

⊕

deposition.

- take place under influence

- climate
- Atmospheric pressure
- human Activity.

Orders of Relief

First

- continental Platforms
- Ocean basin

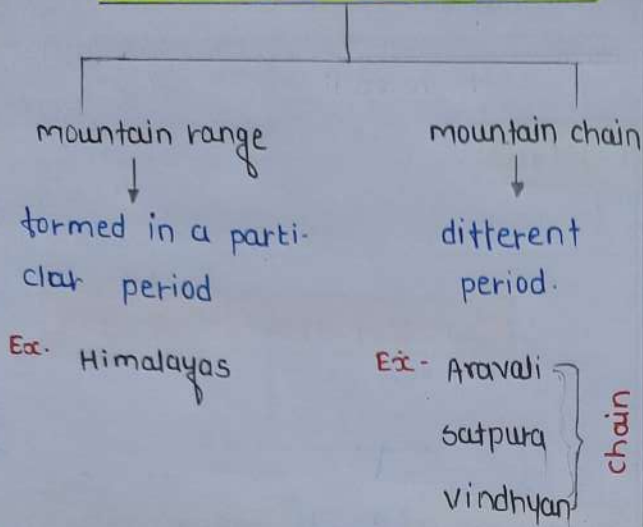
Second

- ⊙ on continents
 - mountains
 - plateaus
 - plains
 - Lowland
- ⊙ In Ocean
 - cont. shelf
 - cont. slope
 - Oceanic ridge
 - Abyssal plain

third

- peaks
- cliff
- valleys
- Hills
- Gorges
- sand dunes
- caves
- Beaches

Mountains - Imp. terms



Grassland

- savana - east Africa
 - campos - Brazil
 - Llanos - Venezuela
 - pampas - Argentina
 - prairies - N. America
 - veld - S. Africa
 - downs - Australia
 - steppes - Eurasia
 - manchuria - China
 - puszta - Hungary
 - Canterbury - NZ
- Tropical
- Temperate

Types of Mountains

- ① Fold mountains
- ② Block mountains
- ③ Residual mountains
- ④ volcanic mountains.

Fold mountains

mountains created due to convergent boundaries

Ex - Mount Everest.

Himalayas - youngest mountain on earth

major fold mountains

Asia

- Himalayas
- Zagros
- Elburz
- Hindukush
- Arkan
- Kunlun
- Altai
- Karakoram

Europe

- Alps
- Ural
- Caucasus
- Pyrenees

Africa

- Atlas
- Drakensberg

N. America

- Rockies
- Appalachians

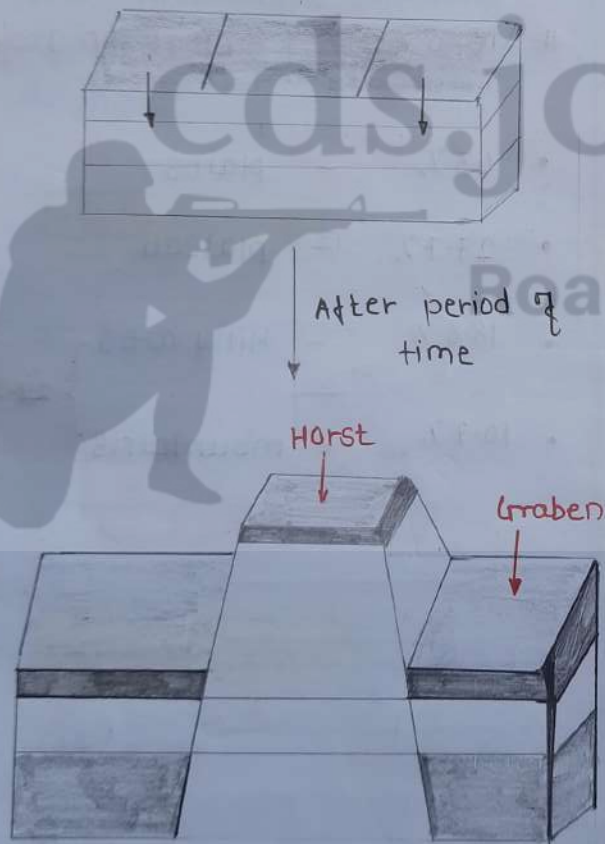
Australia

- Great dividing ranges.

S. America

- Andes

Block Mountains



Largest block m. in World.

USA - Sierra Nevada

India - Satpura, Vindhya

Residual mountains

- old mountains
- upper part become smooth
- Lower height with flat top

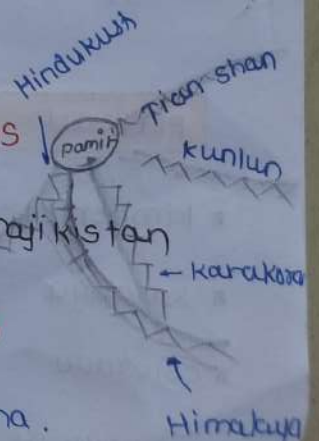
- Ex:
- Aravali
 - Nilgiri
 - Nallamalla hills - TN
 - Palakonda range - AP
 - Henry mount. -
 - Black hill & Mt. Monadnock (USA)

- Mid-Atlantic Ridge
largest landform on the earth

- Urals & Caucasus
divide Europe & Asia

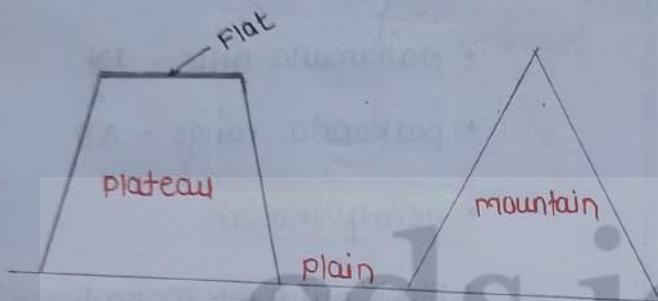
- South Alps mountain range
↓
NZ

- Roof of mountains
↓
Pamir Knot, Tajikistan
- backbone of Asia
↓
Kunlun shan, China.



plateau

- Flat & rough top surface
- steep walls
- Also known - Table land
- 1/3 of continental part.



Types

- Inter-montane plateau
- piedmont
- continental
- volcanic
- Glacial
- Aeolian

plateau

- Kimberley - Aus.
- Katanga - Congo (D.R)
- Colorado - USA
- Mascarene - Indian Ocean

■ Laurentian - Canada

✓ Patagonian - Arg.

■ Massif - France

■ Loess - China

■ Bavarian - Germany

✓ Abyssinian - Ethiopia

✓ Anatolian - Turkey

✓ Iberian - Spain (Europe)

Topographic extent in India

- 43% - plains
 - 27.7% - plateau
 - 18.6% - Hilly area
 - 10.7% - mountains
- } 29.3%

Hydration

- Force of running water wearing down rocks.

Attrition

- River particles collide against each other & break down.

Abrasion

- solid River load striking against rocks & wearing them down.

Downcutting

- vertical erosion



Lateral erosion

- side erosion

width ↑



Headward erosion

origin

stream length
inc. ↑



Landforms due to Exogenic force.

- ① Running water
- ② Ground water
- ③ Glacier
- ④ sea waves
- ⑤ wind

①

Running Water

Erosional Landform

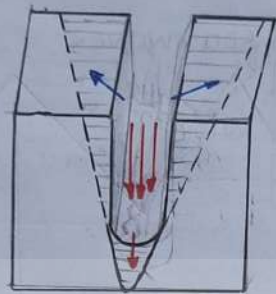
- V-shape valley
- Gorges
- canyon
- Waterfalls
- Rapids
- monadnock
- plunges
- pot holes
- terraces
- Gully / Rills
- peneplain
- Braided channels

Depositional

- Alluvial cones
- Alluvial fans
- flood plain
- Levees
- point Bars
- ox-bow lakes
- Delta
- Estuary

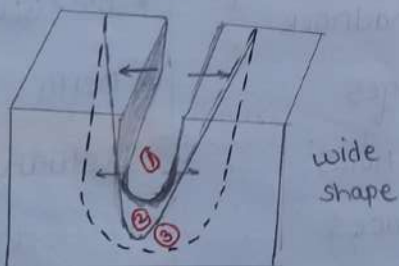
V - Shape valley

- formed - upper course of River
- deep
- Narrow
- steep wall



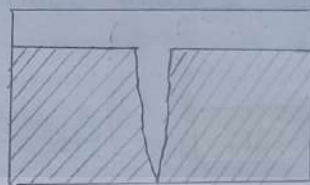
Gorges

- Narrow valley with steep rocks
- blw - hills @ mountains
- due to extensive down cutting erosional force.
- deep & narrow V-shaped valley known - gorges.



canyon

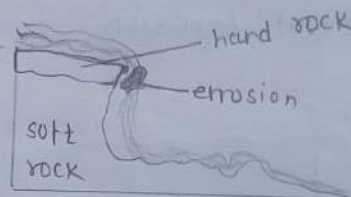
- Deeper version of Gorges with steep slope
- top wider → comp. bottom
- eg: Grand canyon (USA)
(deep)



(canyon)

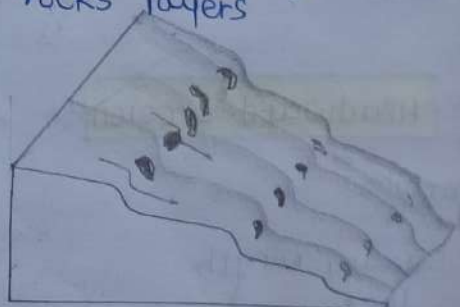
waterfall

youth stage



Rapid

- series of small waterfall
- at surface of
- form a combination of hard & soft rocks layers



Gully / Rills

- formed - water from overland flows down a slope.
- Rills - smaller channels.
- Gully - larger channels.

Peneplain

- formed - stream erosion
- low-relief plain

Depositional Landforms

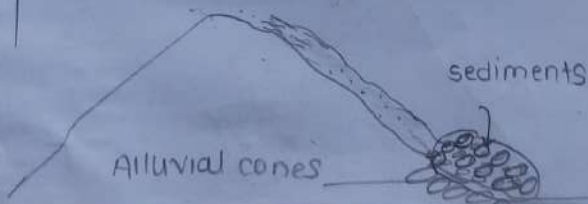
- Alluvial cones
- Alluvial fans
- Flood plain
- Levees
- point Bars
- ox-Bow lakes
- Delta
- Estuary.

• meander
↓
not landforms

↓
River pattern
(way)

Alluvial cones

- streams from higher altitude enter plain - deposited ^(coarse) sediments in cone shape landform.

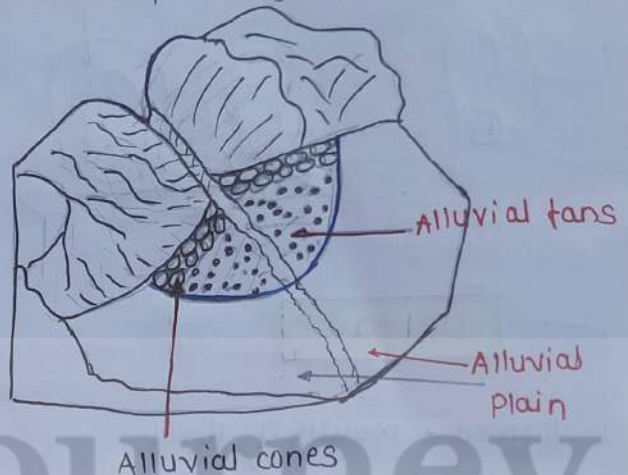


Alluvial fans

- streams - higher Altitude enters plains & fine sediments get deposited.



Fan shape landform



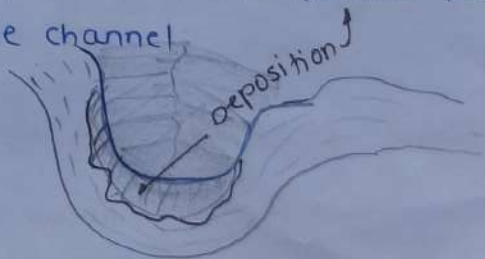
Levees

- bank of rivers formed by deposition of heavy sediments.
- protect flood plain
- PO River (Italy) - huge levees



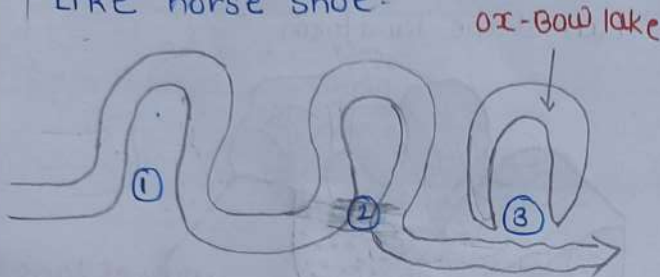
point Bar

- deposition on the convex part of the channel



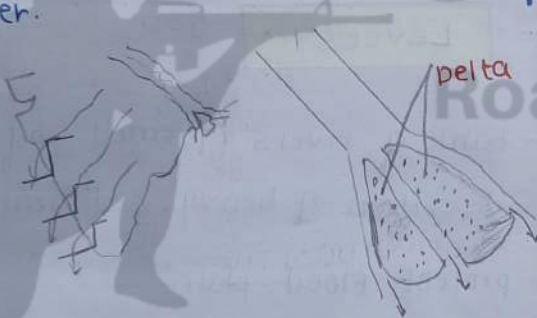
Ox-Bow Lake

- cut off portion of meander
- because of heavy deposition
- Like horse shoe.



Delta

- formed - mouth of River
- large mass of sed. deposit mouth of River.



Estuary

- entrance of River into sea or ocean
- sed. deposited in water

②

Ground Water

↓
karst topography.

erosional

- swallow / sink hole
- valley sink
- Lapies
- caves

Depositional

- stalactite
- stalagmite
- column / pillar

formation of karst topography

- Rocks of limestone or dolomite or gypsum
- sufficient rainfall
- joints in soluble rocks
- land structural relief.

soft rock
ground water erosion

Erosional Landforms

sink hole

- Round or sub round shaped shallow depression
- in surface of limestone.



Valley Sink

Wide & narrow trench formed by the enlargement of sink holes



Caves

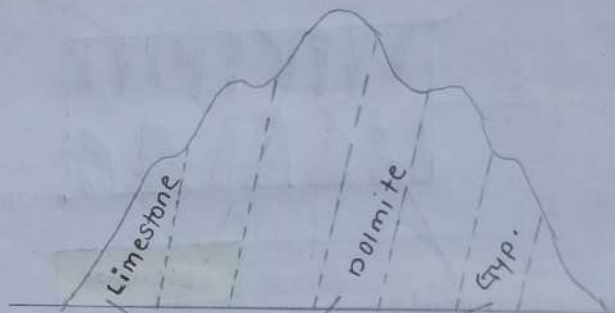
Narrow gaps formed inside surface with alternating beds of rocks with limestones.

Tunnels

caves with openings at both ends.

Lapies

rough & corroded surface
or limestone irregular in shape



erosion



Depositional

Stalactites

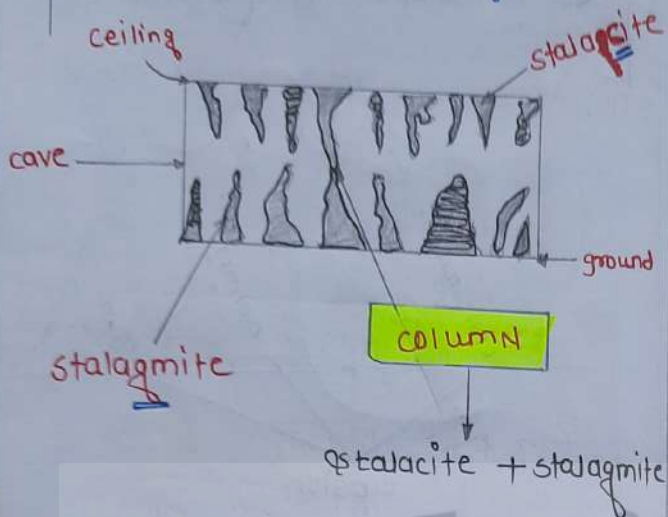
Hanging columns of deposited material on the ceiling of the caves



* CaCO_3 is the chief chemical of deposition on limestone.

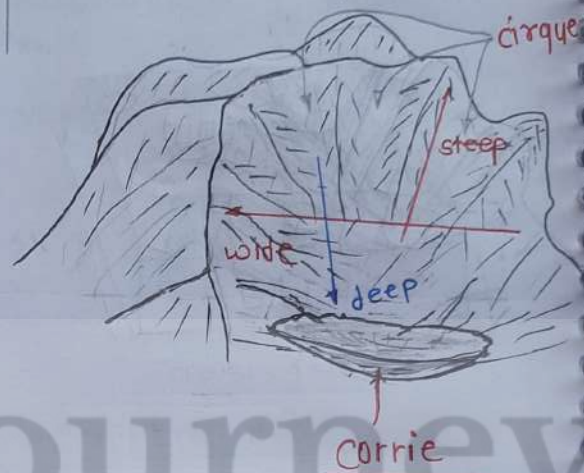
Stalagmite

- Raised columns of the deposited material on the floor of the cave.



Cirque

- Deep long & wide through @ basins.
- very steep high walls at its head as well as sides.



③ Glacial Landforms

erosional	Depositional
• cirque • corrie lake / Tarn lake • Horn • Arêtes • Glacial valleys • Hanging valleys • Fjord • Roche moutonne	• Glacial till • outwash plain • moraines • Eskers • <u>Nunataks</u> Asymmetrical hillocks having one side smooth & other side steep Like - sheep lying posture

Corrie lake

- Tarn lake
- Lake formed at the cirque area after the disappearance of glaciers

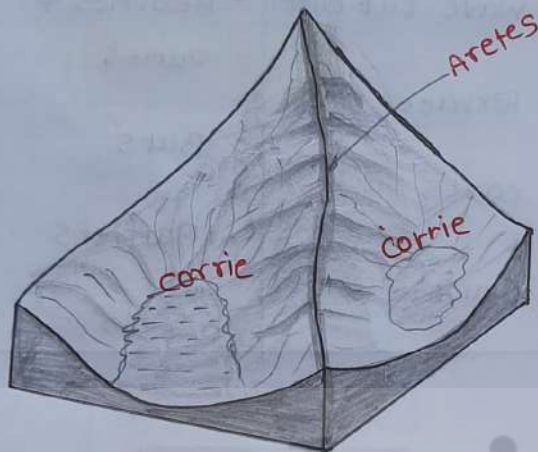
Horn

- formed due to headwards erosion of glacier from ③ or more side.
- sharp pointed & steep sided.



Aretes

- divider between cirque
- it's like a sharp crest.



Fjord

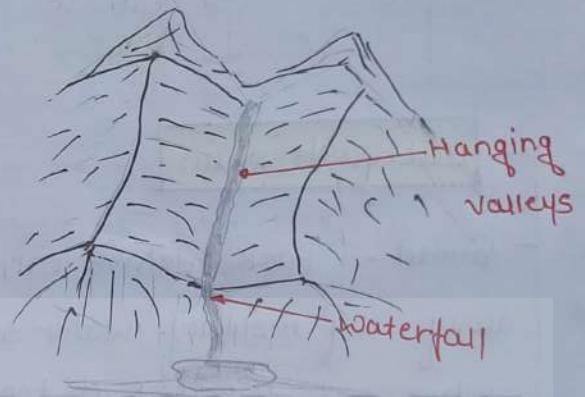
- always connected with sea
- long, deep, narrow body of water that reaches far inland
- Found: Norway, Chile, NZ, Canada, Greenland, US

Glacial Valleys

- U-shaped with broad floors
- steep sides.
- trough

Hanging valleys

- runs above & perpendicular to another glacial valley
- Har Ki Doon



Depositional

Glacial Till

- unsorted sediments deposited directly by glacial ice.

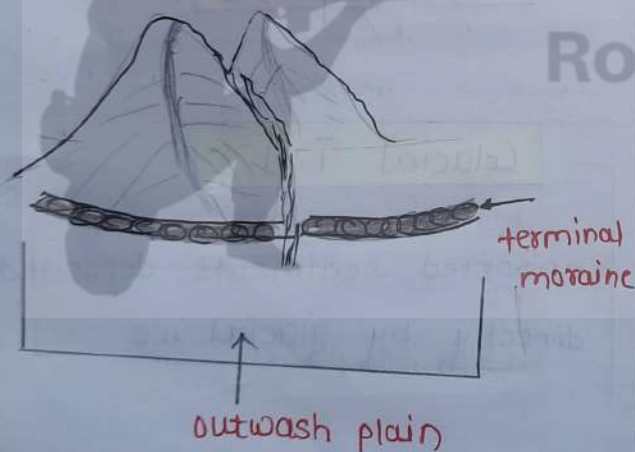


moraines

- sloppy ridges formed by the deposition of eroded & transported sediments carried by melting glaciers.

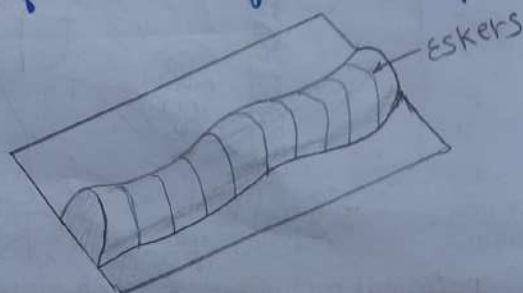
outwash plain

- formed - rock debris carried away by melting water stream when get deposited at foot of mountain.



Eskers

- ridges made of sands & gravels



④

sea wave

erosional	Depositional
• wave cut cliffs • Terraces • caves • stacks	• Beaches & Dunes • Bars • Barriers • spits

Erosional

Wave cut cliff

- steep erosional landform & range - few mt to 30 mt.



Terraces

- flat @ gentle sloppy land-formed by debris of sea cliffs.

Sea caves

Hollow structure at the sea cliffs.

Stacks

Isolated standing rock remnants eroded from all sides.



Depositional

Beaches

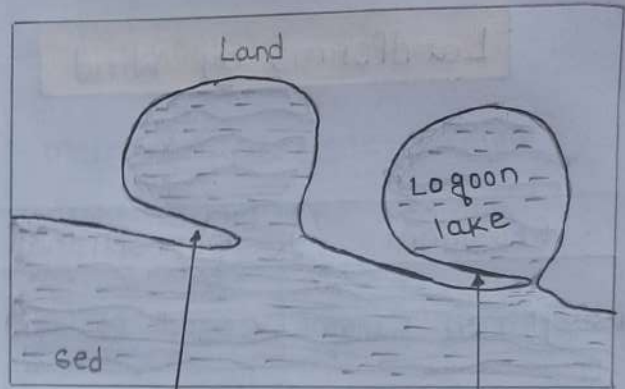
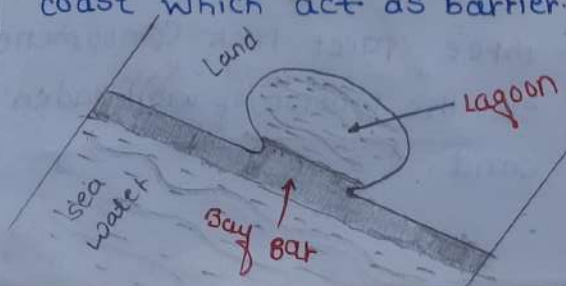
Fine sediments deposited by waves

Sand Dunes

Sand deposited behind the beaches in the form of long parallel ridges.

Bars

A ridge of sand formed in the sea parallel to the coast which act as barrier.



spit

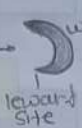
Bar

Sometimes bars are keyed up from one end to another.

Lagoon lakes in Ind.

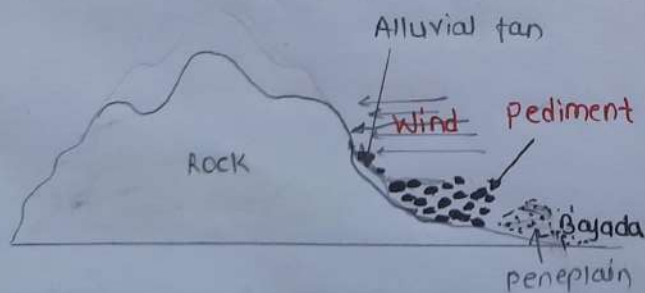
- Chilka - largest (Odisha)
- Pulicat - Second largest (AP) & TN
- Vembanad - Kerala.

Landforms by Wind

Erosional	Depositional
- deflation hollows - Mushroom - yardang - Inselberg - Dreikanter - Mesa - stone lattice - playa - zeugen (pyg 23 II) → table like structure - pediment → *found - areas where glacier have eroded the land-scape.	- sand dunes - Barchan →  - Long dunes - Loess 

pediment

pedineplain
↓
running water
(erosion)



Deflation hollows → qattara depression (egypt)

- shallow depression - caves.
- Genuine dep. - Blowouts
- shallow depression on surface by force of wind.

Mushroom

- formed - when upper rock - hard
- bottom rock - soft
- ↓
- eroded from all side
- take shape of mushroom.



yardang

- steep sided vertically undercut
- rigid (or) rocks eroded by wind
- forms - layers of hard & soft rock (alternate).



Inselberg

- small hill eroded by wind.
- Island mountain in desert areas

Dreikanter

- three facet rock (smoothened by the erosion of wind laden sand).

Mesa

- Spanish - Table
- elevated area of land (or) plateau which isolated from main plateau due to erosion by wind.

Stone Lattice

- Series of tough rocks
- composition of hard & soft structure by high velocity winds.



Depositional

Longitudinal Dunes

- formed - wind direction
 ↓
 long. | linear
- supply of sand deposition is poor.

Loess

- plains (or) plateaus formed by deposition of fine sediments
- Loess - turn fertile when gets water.
- yellow in color due to oxidation

Levees	Lapies	Loess
running water (deposition) Po-River	Ground water (erosion) erosion	wind (deposition)